

The Shift from Traditional Pensions to 401(k)s: Retirement Risks and the Timing of Retirement*

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Abstract

U.S. retirement coverage has shifted from defined benefit (DB) to defined contribution (DC) plans. We build a life-cycle model in which DB and DC plans differ in exposure to longevity and return risk and households respond to the additional risks by saving more and working longer. These two insurance margins interact with one another: additional saving before retirement reduces the need for precautionary labor supply at older ages. The model matches observed differences in retirement timing across plan types and decomposes the effects of the transition into plan structure and generosity, return risk, and longevity risk. Return risk is the main driver of delayed retirement, while longevity risk is the main source of the welfare loss attributable to uncertainty. We then evaluate annuitization as a way to improve the insurance provided by accumulated assets. Allowing retirees to convert assets into actuarially fair annuities raises welfare and induces earlier retirement, with larger gains after the shift toward DC plans.

JEL classification: E21, E24, J26, J32

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1 Introduction

Over the past several decades, U.S. retirement plans have shifted sharply from defined benefit (DB) plans, such as traditional pensions, to defined contribution (DC) plans, such as 401(k)s. This transformation has given individuals greater control over their retirement savings but has also changed both the timing and the risk profile of retirement resources. DB plans generally provide formula based lifetime benefits beginning in retirement, while DC plans accumulate assets in individual accounts that can be accessed and drawn down more flexibly in retirement. This flexibility gives DC participants greater control over when and how retirement wealth is used, but also exposes them more directly to uncertain asset returns and to the risk of outliving their accumulated assets.¹ Following the 1978 Revenue Act², which established the legal foundation for 401(k) plans, DC coverage expanded rapidly and is now the dominant form of private retirement savings.³ Over the same period, labor force participation after age 65 has risen markedly.

This paper studies how differences in longevity and return risk across DB and DC plans affect saving, retirement, and welfare, separately from differences in plan structure and generosity. Households can respond to the additional risks associated with DC plans both by saving more and by working longer. These two responses operate jointly, since additional saving before retirement reduces the need to insure against the same risks by continuing to work at older ages. We show that allowing saving to adjust over the life cycle materially reduces the predicted retirement response to the DB-to-DC transition. We then decompose the effects of the transition by risk source and evaluate annuitization as a way to reduce the underlying risk exposure.

We begin by using detailed panel data from the Health and Retirement Study (HRS) to document differences in retirement timing by plan type. After controlling for detailed household, asset, occupation, health, and other characteristics, participants in DC plans are less likely to retire than those in DB plans at each age throughout their 60s. We then use the data to predict the final retirement age for participants by plan type. Using an instrumental variables approach to address potential endogeneity concerns, we find that participants enrolled in DC plans tend to have significantly higher predicted retirement ages compared to those in DB plans, consistent with an existing empirical literature.

Next, we introduce a stylized model that isolates how longevity and return risks affect saving and retirement timing under DB and DC plans while holding the present value

¹Throughout the paper, we use “return risk” and “uncertain returns” interchangeably, as well as “longevity risk” and “uncertain horizon.”

²Revenue Act of 1978, Pub. L. No. 95-600, 92 Stat. 2763, 2785 (Nov. 6, 1978)

³While the Revenue Act of 1978 did not explicitly create 401(k) plans as we know them today, it laid the legal foundation that allowed these plans to emerge. In 1980, Ted Benna, owner of Pennsylvania benefits consulting company The Johnson Companies, recognized the potential of Section 401(k) to create a formalized retirement savings vehicle. He proposed and implemented the first 401(k) plan, which allowed employees to make pre-tax contributions with matching contributions from employers (Kujawa, 2012; U.S. Department of Labor Employee Benefits Security Administration, 2007).

of each plan fixed. While DB plans bundle longevity insurance and risk pooling into a lifetime benefit stream, DC plans provide individually managed assets that are exposed to uncertain returns and must fund consumption over an uncertain retirement horizon. We demonstrate that individuals in DC plans unambiguously save more, while the impact on retirement timing depends on plan present value. Under moderate plan present values, both risk channels strengthen the incentive to delay retirement for those in DC plans, as working longer provides insurance against the possibility of outliving assets or realizing poor returns. However, when plan benefits are sufficiently generous, earlier access to retirement wealth gives DC participants greater flexibility to retire earlier, while DB participants have less ability to shift consumption across periods. This stylized model serves as an analytically tractable foundation for the quantitative model that follows.

We then construct a quantitative model where tenure in different plan types influences endogenous saving and retirement timing decisions. Agents face an age-dependent death probability each period. Individual and DC plan investments are subject to an uncertain rate of return. Retirement plans are modeled to match key institutional features, with DC plans providing employer contributions while working, and DB plans providing a lifetime benefit that depends on plan tenure and earnings once vesting requirements are met. Agents differ by income group, face unemployment risk, and can transition across plan types when changing jobs. We model retirement plan type as a characteristic of the job offer, with new jobs offering plan types according to income-dependent probabilities disciplined by the data.⁴ We show that the calibrated model reproduces the observed life-cycle patterns of wealth accumulation and retirement and closely matches the observed difference in retirement timing between DB and DC participants.

With the calibrated model, we quantify the effects of the shift from DB to DC plans observed from 1992 through 2018. In the data, labor force participation at age 65 increased by approximately 18.5 percentage points over this period.⁵ The shift from DB to DC plans alone in the model raises participation at age 65 by approximately 13.7 percentage points. Thus, in the calibrated economy, the additional risks faced by DC participants outweigh the flexibility provided by earlier access to retirement wealth. The model also has implications for saving incentives: relative to those whose primary plan is a DB plan, those relying primarily on DC plans accumulate more assets before retirement, but consume a larger portion of their assets during retirement. Consequently, those primarily in DC plans who survive to age 95 have lower average wealth, having depleted a greater share of their savings in retirement. Overall, the movement towards DC plans

⁴Consistent with this assumption, U.S. General Accounting Office (1996) reports that only 3.0% of private employers sponsored both DB and DC plans in 1992, with this share falling to 2.6% in 1993. Even these figures overstate the prevalence of direct worker choice, as the report could not determine whether the two plan types were actually offered to the same employees when both were sponsored.

⁵We show that this data pattern is consistent across various demographic groups, with the increase in participation at age 65 by men alone over this time period being 18.3 percentage points.

and away from DB plans increases the average retirement age and lowers welfare.

An important difference from the existing literature studying the DB to DC plan transition is that our counterfactual allows saving to respond before age 55. Previous work studying this transition reassigns a subset of DB participants to DC coverage at age 55 while holding their previously accumulated assets and other pre-55 states fixed (Bairoliya, 2019). In our full model, the DB-to-DC transition increases participation among ages 65-69 by 4.40%. When we instead remove the effects of the transition on saving before age 55, participation increases by 5.81%, compared with 5.37% in Bairoliya (2019). Thus, allowing households to accumulate additional assets younger reduces the need to respond to greater risk by working longer at older ages. This comparison highlights the importance of allowing saving and retirement decisions to respond jointly.

To clarify which forces drive our results, we decompose the full-model counterfactual effects into those attributable to differences in plan structure and generosity alone and effects due to additional risk exposure under DC plans. The decomposition shows that delayed retirement at age 65 is driven primarily by greater risk exposure. While the full-model transition towards DC plans reduces the share retired by 65 by 13.7 percentage points, 9.8 percentage points are attributable to added uncertainty and 3.9 percentage points to plan-structure/present-value differences alone. Within the uncertainty component, return risk accounts for essentially all of the reduction in retirement by 65. In contrast, welfare losses are split across risk and plan-structure/present-value components. Greater exposure to the two risk channels explains roughly one-third of the decline in welfare, with longevity risk as the primary driver of the welfare loss attributable to uncertainty. Thus, even though households respond to the additional risks associated with DC plans by saving more and working longer, these responses do not fully protect them from the welfare costs of retirement risk.

This remaining exposure to longevity and return risk motivates our annuity counterfactual. Annuities allow households to convert accumulated assets into guaranteed lifetime income, reducing the risk of outliving their savings. However, voluntary annuitization remains limited in practice. Hosseini (2015) finds that less than 1% of retirement wealth is annuitized, pointing to adverse selection pushing prices above actuarially fair levels. Similarly, Verani and Yu (2024) shows that the supply of annuities is also constrained by interest rate risk.⁶ Given these frictions, we evaluate a counterfactual policy that allows retirees to annuitize any fraction of their accumulated assets at actuarially fair rates. Annuitization reduces both longevity risk and exposure to uncertain returns after retirement, raises welfare, and reduces the need to work longer as insurance against retirement risk. These effects are larger after the transition toward DC plans, when fewer households receive annuity-like income through DB plans.

⁶This challenge was recently focused on by the 2022 SECURE 2.0 Act, which aimed to reduce barriers to in-plan lifetime income options and expand Qualifying Longevity Annuity Contracts (QLACs).

The remainder of this paper is outlined as follows: Section 2 discusses the related literature and the contribution of this paper, Section 3 discusses our empirical motivation, Section 4 presents the illustrative model, Section 5 presents the quantitative model, Section 6 presents model calibration and counterfactual results, and Section 7 concludes.

2 Related Literature

The contribution of this paper is to show how differences in retirement-plan design affect saving and retirement decisions over the life cycle. DB and DC plans differ not only in exposure to return and longevity risk, but also in the timing and flexibility of retirement resources. DC plans provide earlier access to retirement wealth and greater flexibility over how those resources are used, but expose individuals to greater return and longevity risk than DB plans. Households can respond to this additional risk both by saving more and by working longer. These responses operate jointly, since additional asset accumulation before retirement reduces the need to insure against the same risks by continuing to work at older ages. We embed these channels in a calibrated life-cycle model and decompose the effects of the DB-to-DC transition into differences in plan structure and generosity, return risk, and longevity risk. We then evaluate whether allowing retirees to convert accumulated assets into actuarially fair annuities can reduce the underlying risk exposure and its effects on retirement and welfare.

This paper contributes to the empirical discussion on how variations in retirement-plan incentives influence retirement timing. Classic work emphasized that DB plans can generate sharp retirement spikes through age-dependent accrual and option values, encouraging retirement at specific ages (Stock and Wise, 1990). Related work documents that heterogeneity in DB plan benefits is quantitatively important for explaining labor force participation patterns from 1950 into the 1980s (Samwick, 1998). A more recent empirical literature compares retirement behavior under DB and DC coverage. Munnell, Cahill and Jivan (2003), Munnell, Triest and Jivan (2004), Friedberg and Webb (2005), and Manchester (2010) find that individuals retire earlier with DB coverage than with DC coverage after controlling for a rich set of observables. We complement these empirical findings by addressing potential selection into plan types. In our analysis, we instrument for plan coverage using unionization status, controlling for occupation, industry, region, and other characteristics. For workers in the same industry, occupation, and region, union coverage is strongly linked to DB plan availability. This approach helps isolate differences due to retirement plan type rather than unobservable preferences correlated with employment under each plan type. Because retirement plan type is typically attached to the job, the relevant selection margin is primarily sorting across employers, occupations, and sectors rather than a direct within-job choice between DB and DC plans.

On the modeling side, we build on quantitative life-cycle models of saving and retire-

ment (French, 2005; Blau, 2008; De Nardi, French and Jones, 2010; French and Jones, 2011; Fan, Seshadri and Taber, 2024) and on work studying the historical transition from DB to DC plans in the U.S. (Heiland and Li, 2012; Bairoliya, 2019). The closest paper is Bairoliya (2019), which develops a detailed model of retirement, saving, Social Security, health insurance, and DB and DC pension incentives beginning at age 55. Bairoliya (2019) studies the change in pension composition by randomly reassigning a subset of DB participants to DC coverage at age 55 while holding their other initial conditions and wage process fixed. This isolates differences in retirement incentives after age 55, but does not allow the change in plan coverage to affect saving and wealth accumulation earlier in the life cycle.

Our counterfactual instead allows the transition from DB to DC plans to affect decisions beginning at age 30. This distinction is important because saving more and working longer provide two ways for households to respond to the additional risks associated with DC plans. In our full model, the DB-to-DC transition increases participation among ages 65–69 by 4.40%. When we instead follow the age-55 reassignment approach in Bairoliya (2019) and remove the effects of the transition on saving before age 55, participation increases by 5.81%, close to the 5.37% increase in Bairoliya (2019). Thus, allowing households to accumulate additional assets before retirement reduces the need to respond to greater risk by working longer at older ages. This comparison highlights the importance of allowing saving and retirement decisions to respond jointly over the life cycle.

The models also differ in how retirement-plan histories evolve. In Bairoliya (2019), pension type is a permanent characteristic beginning at age 55. In our model, workers can move between plan types when changing jobs, so plan histories and DB tenure evolve endogenously over the life cycle. Workers who leave a DB-covered job retain benefits associated with their accumulated DB tenure, while additional DB tenure raises future benefits until the maximum tenure is reached. We also account for DB vesting requirements and include expected spousal DB survivor benefits in the value of bequests, while accumulated DC wealth is directly bequeathable. Finally, Bairoliya (2019) randomly reassigns the same fraction of DB participants in the pension-shift counterfactual, whereas our plan-offer probabilities vary by income group. In the data, the historical shift from DB toward DC coverage was substantially larger for middle- and higher-income workers. These features allow the transition to affect both the resources households accumulate before retirement and the retirement incentives they face at older ages.

Relative to the existing literature, we also make the risk properties of retirement-plan design central. Holding the present-value cost of benefits fixed, we show how greater exposure to return and longevity risk under DC plans affects both saving and retirement incentives. We then use the quantitative model to separate the effects of plan structure and generosity from the additional uncertainty faced under DC plans, and further decompose the uncertainty component into return risk, longevity risk, and their interac-

tion. This decomposition allows us to identify which features of the transition are most important for retirement, asset accumulation, and welfare.

Our paper connects to the literature on the relationship between retirement-plan wealth and private saving. Using DB plan reforms across various countries, prior work finds that reductions in DB pension wealth raise private saving, although the magnitude varies across settings (Attanasio and Brugiavini, 2003; Attanasio and Rohwedder, 2003; Engelhardt and Kumar, 2011; Alessie, Angelini and van Santen, 2013; Lachowska and Myck, 2018). Our model is broadly consistent with this evidence, but emphasizes the interaction between this saving response and retirement. When households accumulate additional assets in response to greater retirement risk, they have less need to respond to the same risk by working longer. Accounting for both margins is therefore important for measuring the retirement effects of changes in retirement-plan design.

Finally, our paper connects the DB-to-DC transition to the literature on annuitization. Although households can partly insure against retirement risk by accumulating additional assets, these assets remain exposed to uncertain returns and must fund consumption over an uncertain retirement horizon. At the same time, voluntary annuitization remains extremely limited because of well-documented market frictions (Hosseini, 2015; Verani and Yu, 2024). Rather than modeling these frictions explicitly, we use our structural framework to evaluate a counterfactual in which retirees can convert accumulated assets into actuarially fair annuities. Annuitization reduces longevity risk and exposure to uncertain returns after retirement, making accumulated assets a more effective source of retirement insurance. We find that access to annuitization raises welfare and induces earlier retirement, with larger gains after the transition toward DC plans.

3 Empirical Motivation

In this section, we document three empirical patterns that motivate the analysis. First, the composition of retirement plan coverage has shifted substantially from DB toward DC plans since the early 1990s. Over the same period, labor force participation at older ages increased markedly. We view these concurrent trends as motivation for studying how differences in retirement plan design affect retirement behavior, rather than as evidence of a causal relationship between the two aggregate trends. Second, conditional on detailed individual, household, health, and employment characteristics, individuals enrolled in DB plans are more likely to transition into retirement than those enrolled in DC plans. Third, individuals whose primary retirement plan is a DC plan have significantly higher predicted final retirement ages than those whose primary plan is a DB plan.

3.1 Data and Retirement Plan Classification

Our primary data source is the Health and Retirement Study (HRS) from 1992 to 2020, which provides longitudinal information on retirement plan coverage, employment, retirement, wealth, health, and household characteristics.⁷ The HRS primarily follows individuals over age 50 and their spouses, regardless of the spouse’s age. Interviews began in 1992 and are generally conducted every two years, with additional cohorts entering the survey over time. The longitudinal structure of the HRS allows us to observe retirement plan coverage and labor force status repeatedly for the same individuals as they approach and enter retirement. Moreover, it is designed to link spouse and household information, allowing us to account for household-level effects in our analysis.

The original HRS dataset consists of 280,343 individual year observations from 42,405 unique respondents with valid survey responses. We exclude respondents with missing information on age, race, or census region of residence. Because our analysis focuses on retirement decisions, we also exclude individuals who were already retired at their first observed interview. We further require valid information on labor force status, retirement plan coverage in the current main job, retirement status, self reported health, union coverage, and health, long term care, and life insurance coverage. Table 13 in Appendix B.1 reports the sample selection procedure in detail. The resulting sample contains 219,887 observations from 32,791 unique individuals, representing more than 78 percent of the observations in the original HRS sample.

During each interview, employed respondents report detailed information about retirement plans associated with their current jobs. We focus on the respondent’s current main job and classify retirement coverage into four broad categories: no retirement plan, DB plan, DC plan, and hybrid plan. Table 1 summarizes the detailed retirement plan types reported in the HRS and their corresponding classifications. DB plans provide formula based benefits that are generally paid as a retirement income stream, whereas DC plans accumulate assets in individual accounts whose eventual value depends on contributions and investment returns. Hybrid plans combine features of DB and DC arrangements.

Table 1: Categorization of Detailed Retirement Plans

Types of Retirement Plans	Detailed Types
No Plan	No retirement plans or only health plan mentioned
Defined Benefit Plan	Defined benefit plan
Defined Contribution Plan	Defined contribution plan; 401k plan; 401a plan; 403b plan; 457 plan; thrift/savings plan (tsp); profit-sharing plan; employee stock ownership; money purchase plan; employee stock purchase; sep or simple plan
Hybrid	Combination plan; cash balance plan

⁷We use the RAND HRS Longitudinal File 2020 (V2).

Although retirement plan type can change over an individual’s career, such changes are relatively uncommon in the HRS sample. Approximately 86 percent of participants never change retirement plan type during the years in which they are observed working. Appendix B.2 provides the distribution of the total number of observed plan switches. This persistence allows retirement plan histories in the HRS to provide useful information about the pension environment individuals face as they approach retirement.

We supplement the HRS with data from the Current Population Survey (CPS) to document broader changes in retirement and labor force participation at older ages. The HRS is our primary data source because it provides detailed longitudinal information on retirement plan coverage together with individual employment histories, wealth, health, and other household characteristics. The CPS, in contrast, provides large repeated cross sections that allow us to trace aggregate retirement and labor force participation patterns over a longer period. We use the CPS to document changes in retirement among individuals aged 65 and older and changes in labor force participation at older ages. These CPS patterns provide aggregate motivation for the analysis, while the HRS allows us to examine how retirement behavior differs across DB and DC plan participants at the individual level.

3.2 Changes in Retirement Plan Coverage and Labor Force Participation

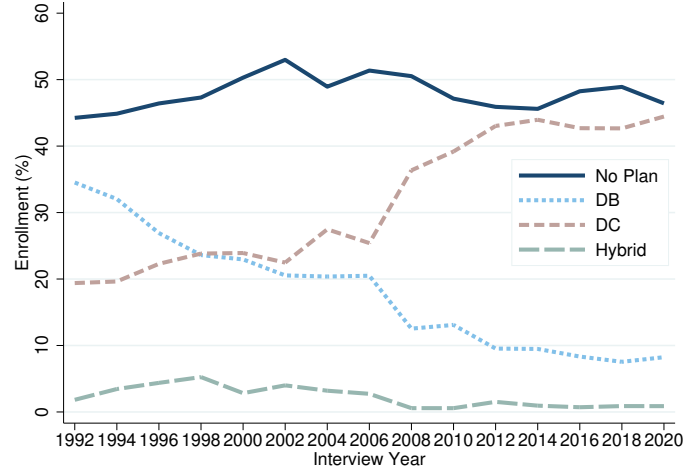
We first document two concurrent changes in the U.S. retirement environment: the transition from DB toward DC coverage and the increase in labor force participation at older ages.

Figure 1 displays retirement plan coverage among HRS participants by interview year. The figure shows a substantial change in the composition of retirement plans beginning in the early 1990s. The shares of participants with no retirement plan or with a hybrid plan remain relatively stable over much of the sample period. In contrast, the share enrolled in DC plans rises substantially, while the share enrolled in DB plans declines. Between the early 1990s and 2020, DC enrollment increases by more than 20 percentage points, accompanied by a decline of similar magnitude in DB enrollment. The data therefore document a pronounced movement away from traditional DB pensions and toward individual account based DC plans.

During the same period, retirement behavior at older ages also changed substantially. Figure 2 documents these patterns using the CPS. Panel (a) reports the probability of being retired among individuals aged 65 and older.⁸ Since the mid 1990s, this probability has declined by approximately 7.8 percentage points. Because changes in the age com-

⁸The data does not allow us to see if respondents report being “retired” before 1994, although we observe their participation status.

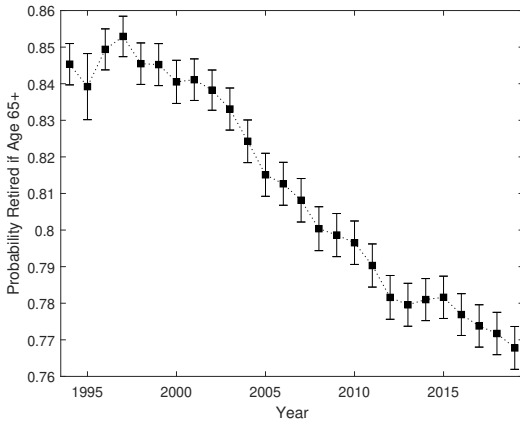
Figure 1: Retirement Plans of Current Main Job in Each Interview Year



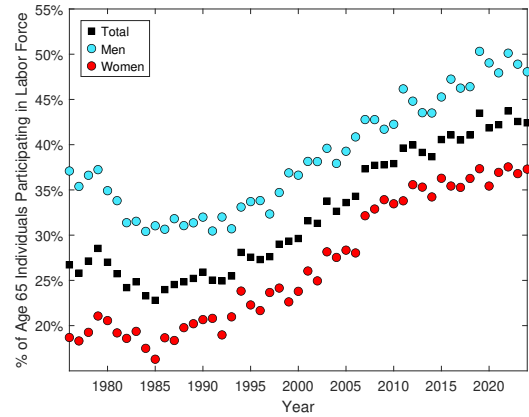
Notes: Data displayed is from the Health and Retirement Study (HRS). The categorization of possible detailed retirement plan types into defined benefit (DB), defined contribution (DC), or hybrid categories is described in Table 1.

position of the population aged 65 and older could contribute to this pattern, Panel (b) focuses on labor force participation at exactly age 65. Participation at age 65 declined modestly through the early and mid 1980s, but began rising around the early 1990s. The increase is evident for both men and women and amounts to approximately 18.5 percentage points for the population as a whole over the period considered. In Appendix C.2, we show that the increase in participation persists when separately analyzing college graduates and non-graduates of either gender.

Figure 2: Decrease in Retirement at Age 65



(a) Retirement Probability if Age 65+



(b) Age 65 Participation Rate

3.3 Differences in Retirement Timing by Plan Type

We next examine whether retirement behavior differs systematically across retirement plan types. We begin with the probability that an employed individual transitions into retirement. Using a logistic regression model, we estimate the probability of transitioning from employment to retirement for individuals aged 60 to 69.

Table 2 reports the estimated effects of retirement plan type under alternative fixed effects specifications. Model (1) includes year fixed effects only, capturing macroeconomic shocks and time-specific factors affecting retirement decisions. Model (2) replaces year fixed effects with birth cohort fixed effects to account for potential generational differences in retirement behavior and pension coverage. Model (3) includes both year and cohort fixed effects, thereby controlling for aggregate time trends and cohort-specific characteristics. In all specifications, we control for demographic characteristics, income measures, self-reported health conditions, and coverage of health, long-term care, or life insurance. We further include controls for educational degree, occupation, industry, and region of residence. For brevity, the coefficients on these control variables are not reported.

Across all three specifications, DC enrollment is associated with a significantly lower probability of transitioning into retirement relative to DB enrollment. The estimated coefficient on DC coverage is also highly stable across specifications. For example, the DC coefficient changes only from -0.417 in the specification with year fixed effects to -0.430 when both year and cohort fixed effects are included. The stability suggests our results are not driven by conflation of cohort and time effects.

We conduct several robustness exercises to assess whether these differences reflect other characteristics correlated with retirement plan type. First, we allow the relationship between plan type and retirement transitions to vary across birth cohorts. The interactions between retirement plan type and cohort indicators are jointly insignificant, providing no evidence that the DB and DC differential changes systematically across generations. Second, because DB plans are substantially more common in the public sector, we reestimate the model after excluding individuals employed in Public Administration. The estimated relationship between DC enrollment and retirement transitions remains similar. These results are reported in Appendix C.1.

We next examine differences in final retirement age between DB and DC participants. We estimate the following specification:

$$RetireAge_i = \alpha + \beta' RetirePlan_i + \gamma' X_i + \epsilon_i \quad (1)$$

where $RetireAge_i$ denotes the final retirement age of individual i . Because the retirement age represents the individual's last exit from the labor force, we use the most frequent (primary) retirement plan observed for each person rather than time-varying plan enrollment over the life cycle. The variable $RetirePlan_i$ includes indicators for DC plan

Table 2: Employment to Retirement Transitions Among Ages 60-69

	(1) Year FE		(2) Cohort FE		(3) Year + Cohort FE	
	Estimates	S.E.	Estimates	S.E.	Estimates	S.E.
Retirement Plans						
DC	-0.417***	0.062	-0.409***	0.062	-0.430***	0.063
Hybrid	-0.054	0.135	-0.103	0.134	-0.083	0.135
Demographic Controls						
Age	3.298***	0.447	3.174***	0.451	3.292***	0.457
Age squared	-0.025***	0.004	-0.025***	0.004	-0.025***	0.004
Female	0.171*	0.072	0.156*	0.072	0.169*	0.073
Never married	-1.317	1.067	-1.329	1.080	-1.308	1.074
Race: Black/African American	-0.201*	0.100	-0.233*	0.101	-0.231*	0.102
Race: Other	-0.326*	0.155	-0.326*	0.156	-0.338*	0.157
Income Controls						
Hourly wage	0.004*	0.002	0.004*	0.002	0.004*	0.002
Hourly wage squared	-0.000	0.000	-0.000	0.000	-0.000	0.000
Income from retirement plans	0.000*	0.000	0.000*	0.000	0.000	0.000
Income from retirement plans squared	-0.000	0.000	-0.000	0.000	-0.000	0.000
Income from social security	0.000	0.000	0.000	0.000	0.000	0.000
Social security income squared	-0.000	0.000	0.000	0.000	-0.000	0.000
Household wealth (incl. house)	-0.000	0.000	-0.000	0.000	-0.000	0.000
Health Controls						
Self-reported health: very good	0.183*	0.088	0.194*	0.088	0.186*	0.088
Self-reported health: good	0.399***	0.090	0.401***	0.091	0.406***	0.091
Self-reported health: fair	0.533***	0.116	0.560***	0.116	0.538***	0.117
Self-reported health: poor	0.553*	0.233	0.576*	0.231	0.529*	0.231
Spouse health: very good	0.053	0.091	0.049	0.091	0.042	0.091
Spouse health: good	-0.042	0.093	-0.047	0.093	-0.056	0.093
Spouse health: fair	-0.161	0.111	-0.156	0.111	-0.168	0.111
Spouse health: poor	0.136	0.146	0.124	0.148	0.123	0.148
Insurance Coverage Controls						
Health insurance coverage	0.167	0.254	0.177	0.254	0.197	0.255
Long-term care insurance coverage	0.134	0.091	0.134	0.091	0.144	0.092
Life insurance coverage	-0.036	0.091	-0.023	0.092	-0.035	0.092
Constant	-108.395***	14.069	-103.790***	14.209	-108.004***	14.413
Observations	7,697		7,739		7,687	

Notes: The table shows the results of a logistic regression using HRS sample. Estimations are adjusted by the HRS personal-level weights. The regression also controls for the participant's educational degree, occupation, industry, as well as region and year fixed effects. The reference category is married White men enrolled in a DB retirement plan in their current main job, with excellent self-reported health, no health insurance, no long-term care insurance, and no life insurance. Their spouses also report excellent health. Robust standard errors are presented. Stars denote statistical significance * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

enrollment (with DB as the omitted category), and $\mathbf{X}_i$ is a vector of control variables, including household wealth, pre-retirement earnings and its square, wage and its square, health status, gender, race, education, marital status, region, health insurance, long-term care insurance, life insurance coverage, pre-retirement occupation, industry, and retirement year fixed effects.⁹

A potential concern is that retirement plan coverage may be endogenous. Individuals with different preferences for retirement timing may sort into employers or occupations that offer different types of retirement plans. Such sorting could cause the estimated

⁹In the data, around 20% of individuals receive retirement income from multiple retirement plans. To minimize potential bias arising from multiple benefit types, we restrict the sample to those who report receiving income from only one plan type. We assume this income originates from the individual's primary retirement plan, identified as the plan type most frequently observed during their career after age 50, since the sample primarily consists of individuals over 50.

relationship between retirement plan type and retirement age to reflect unobserved worker characteristics rather than differences in retirement plan incentives. To address potential endogeneity in retirement plan choice, we instrument the retirement plan dummies using union status. To address this concern, we supplement the ordinary least squares estimates with an instrumental variables specification that uses pre retirement union coverage as an instrument for DC enrollment. We find the instrument is strongly related to pension type in the first stage: unionized workers are 25 percentage points less likely to be enrolled in DC plans ($t = -7.07$). The excluded-instrument F-statistic equals 49.97, well above conventional weak-instrument thresholds (Stock–Yogo 10% critical value = 16.38). These results indicate that the instrument is strong and the model is well identified. In Appendix C.1, Table 16 provides a detailed comparison between OLS and IV estimates.

The relevance of union coverage for pension type is also consistent with institutional evidence. Appendix C.3 presents evidence from a difference-in-differences design exploiting the staggered adoption of Right-to-Work (RTW) laws. Because RTW laws reduce union coverage, they generate exogenous variation in DB plan enrollment. We find that RTW adoption leads to a significant decline in DB participation, reinforcing the link between union status and pension type.

Regarding the exclusion restriction, union coverage may affect retirement timing through institutional channels other than pension design, such as retiree health benefits, job protection, or workplace flexibility. To mitigate these concerns, we use a rich set of controls, including household wealth, pre-retirement earnings, wages and their square, self-reported health status, health insurance coverage, long-term care insurance coverage, life insurance coverage, pre-retirement occupation, industry, region, and retirement year fixed effects.

While the exclusion restriction cannot be tested directly in a just-identified setting, several pieces of evidence support the relevance of pension structure as an important channel. First, union coverage strongly predicts retirement plan type in the first stage. Second, the first-stage relationship is consistent with institutional evidence from Right-to-Work reforms, which reduced union coverage and DB participation. Third, the IV estimates are robust to excluding public-sector workers, suggesting that the results are not driven solely by the concentration of DB plans in government employment.

Table 3 summarizes the IV-based predicted final retirement ages by retirement plan type. The estimates imply a predicted retirement age of approximately 61.4 for DB participants and 66.7 for DC participants. Thus, the HRS evidence consistently indicates later retirement among participants whose primary retirement plan is a DC plan. The transition regressions show that DC participants are less likely to retire at a given age, while the final retirement age analysis shows that they remain in the labor force longer on average.

These empirical patterns motivate the models that follow. The data establish a robust

Table 3: Predicted Retirement Age by Retirement Plans

	Predicted Age	95% Confidence Interval
DB	61.439*** (0.525)	[60.410, 62.467]
DC	66.677*** (0.908)	[64.898, 68.457]

Notes: The table presents the predicted retirement ages for participants enrolled in Defined Benefit (DB) and Defined Contribution (DC) retirement plans, estimated using instrumental variable regression, based on HRS sample. The main regression model is specified in Equation (1). Robust standard errors are reported in parentheses. Statistical significance is indicated by asterisks: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

difference in retirement timing between DB and DC participants, but they do not by themselves identify which institutional features of the two plans generate this difference. DB and DC plans differ in the timing and flexibility of retirement resources as well as in exposure to uncertain investment returns and an uncertain planning horizon. Section 4 uses a stylized model to isolate how return risk and uncertain horizon affect retirement incentives under the two plans, and Sections 5 and 6 quantify these mechanisms in a richer life cycle model.

4 Illustrative Model

This section presents a simple model to isolate how differences between defined contribution (DC) and defined benefit (DB) retirement plans affect savings and retirement timing, holding fixed the present-value of each plan. The model is used to build intuition ahead of the quantitative model in Section 5. It highlights two key channels: (i) an uncertain planning horizon channel, because agents may not survive to the second period, and (ii) an uncertain rate of return channel, because DC retirement resources are exposed to investment risk while DB benefits are guaranteed. When plan present value is modest, both channels make early retirement relatively more attractive under DB plans, while individuals with DC plans have a stronger incentive to continue working and saving.

However, the effect of plan type on retirement timing is ambiguous. DC participants receive retirement wealth earlier and can use it to finance either current consumption or future retirement. When DC wealth is sufficiently large, this flexibility can outweigh the insurance provided by DB plans, allowing DC participants to retire earlier in extreme cases. Thus, the model shows why DC plans generally delay retirement through greater exposure to risk, while also identifying conditions under which their earlier access to retirement wealth can reverse this result.

4.1 Setup and Timing

Consider a two-period model with the following timeline. In period one, agents enter with initial assets a and retirement plan type $i \in \{B, C\}$. $i = B$ denotes a defined benefit plan and $i = C$ denotes a defined contribution plan. Agents receive first-period retirement plan benefit f^i , where $f^B = 0$ and $f^C \geq 0$.¹⁰ Agents then decide whether to retire at the start of the first period. Those who continue to work earn income $y > 0$, while those who retire receive a utility benefit of leisure $\ell_R > 0$. After making the first-period retirement decision, agents decide how much to save for the next period. Savings earn an ex ante unknown return, which takes the value r_z with probability p_z , for each $z \in Z$. Agents die and exit the model with probability $\nu \in (0, 1)$ before the start of the second period. All agents retire in the second period, realize the returns on previous savings, and consume their remaining resources. In the second period, a retirement plan benefit x^i is paid, where $x^B \geq 0$ and $x^C = 0$. Everyone exits the model after the second period.

Let $V_1(a)$ denote the value of an agent in period one with initial assets a and plan type i . The agent chooses at the start of period one to retire ($R = 1$) or not retire ($R = 0$). Assume agents get utility $u(c)$ from consumption c , with $u'(c) > 0$ and $u''(c) < 0$.

$$V_1^i(a) = \max_{R \in \{0,1\}, s \in [0, y(1-R) + f^i + a]} \{u((1-R)y + f^i + a - s) + R\ell_R + \beta(1-\nu)\mathbb{E}[V_2^i(s)]\}$$

The agent's budget constraint in the first period is as follows. Their consumption c and savings s equal any income from work y , existing assets a , and any first-period retirement plan payment f^i . Therefore, first-period consumption is $c_1^i = (1-R)y + f^i + a - s$.

In the second period, all agents retire and receive leisure value ℓ_R . They consume any remaining assets a plus any second-period retirement plan income x^i . Given that agents do not know the value of the risky rate of return r_z when making their first-period savings decision, the expected period-two value given savings s in period one is as follows.

$$\mathbb{E}[V_2^i(s)] = \ell_R + \sum_{z \in Z} p_z u(x^i + s(1 + r_z))$$

¹⁰In this setup, f^i is paid before agents decide whether to retire in the first period. With this assumption, along with the assumption that all agents retire in the second period, the agents' decisions cannot alter their expected plan value. This generates a simple setup where we can hold the present value cost of the plan types fixed and compare the incentives induced by early DC plan benefits versus later DB plan benefits.

4.2 Savings

First, we study how the DB benefit x^B and DC benefit f^C affect optimal savings. Conditional on the period-one retirement choice R , the savings first-order condition is:

$$u'((1-R)y + f^i + a - s) = \beta(1-\nu) \sum_{z \in Z} [p_z(1+r_z)u'(x^i + s(1+r_z))].$$

Agents equate the marginal cost of saving one more unit today with the discounted expected return-weighted marginal utility benefit in the next period. An increase in f^i reduces the marginal cost of saving in period one, because it directly increases the pool of resources available in this period. Therefore, an increase in f^i raises optimal savings. Alternatively, an increase in x^i reduces the benefit of having an additional dollar saved for period two, because x^i directly increases period-two consumption. Therefore, an increase in x^i decreases optimal savings. Beyond this intuitive explanation, Appendix A.1 formally shows that optimal savings are strictly increasing in f^i and strictly decreasing in x^i .

Let c_1^{iR} denote consumption in period one given retirement choice $R \in \{0, 1\}$ and retirement plan $i \in \{B, C\}$. Similarly, let $c_2^{iR}(r_z)$ denote the analogous consumption for period two, given realized return r_z . With this notation, the Euler equation regarding optimal saving can be written more compactly as

$$u'(c_1^{iR}) = \beta(1-\nu) \sum_{z \in Z} [p_z(1+r_z)u'(c_2^{iR}(r_z))]. \quad (2)$$

4.3 Retirement Timing

Now consider how increasing the present value cost φ^i of providing each type of retirement plan $i \in \{B, C\}$ will affect retirement choices. The present value cost of providing the DC plan benefit paid in period one is simply $\varphi^C = f^C$. To cover the expected DB plan benefit in period two, a firm must set aside an amount φ^B so that $\varphi^B \sum_{z \in Z} p_z(1+r_z) = x^B(1-\nu)$. The two plan types will have the same present-value cost when $\varphi^B = \varphi^C = \frac{x(1-\nu)}{\sum_{z \in Z} p_z(1+r_z)}$. Given the present value cost of providing the plan φ^B , the DB plan benefit in period two is $x^B = \varphi^B \left(\frac{1}{1-\nu}\right) \sum_{z \in Z} p_z(1+r_z)$. Going forward, we will consider how retirement choices respond to changes in the present value plan cost when $\varphi^C = \varphi^B = \varphi$.

Letting $V_1^{i0}(a)$ denote the first-period value if continuing to work ($R = 0$) with plan type i , and $V_1^{i1}(a)$ denote the first-period value if choosing to retire ($R = 1$), consider the value $V_1^{i1}(a) - V_1^{i0}(a) \equiv \Delta V^i(a)$. If $\Delta V^i(a) > 0$, the agent will optimally choose to retire in the first period, while if $\Delta V^i(a) < 0$ they will continue working. Consider how

an increase in the present value cost of either plan type affects $\Delta V^i(a)$.

$$\begin{aligned}\Delta V^i(a) &= \ell_R + u(f^i + a - s^{i1}) - u(y + f^i + a - s^{i0}) \\ &\quad + \beta(1 - \nu) \sum_{z \in Z} p_z [u(x^i + s^{i1}(1 + r_z)) - u(x^i + s^{i0}(1 + r_z))]\end{aligned}$$

First, consider an agent with a DC plan where $f^i = f^C \geq 0$ and $x^C = 0$. We will examine how an increase in the present value plan cost $\varphi = f^C$ affects the expected gain of period one retirement.¹¹ After substituting in the first-order condition regarding optimal savings given by equation (2) and simplifying, we get

$$\frac{\partial \Delta V^C}{\partial \varphi} = \beta(1 - \nu) \sum_{z \in Z} p_z(1 + r_z) [u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))]. \quad (3)$$

Now consider an agent with a DB plan where $f^B = 0$ and $x^B \geq 0$. To be consistent with the DC plan case, we will consider a change in the present value cost φ of the DB plan. Recall, given x^B , $\varphi = \frac{x^B(1-\nu)}{\sum_{z \in Z} p_z(1+r_z)}$, so that $x^B = \varphi \left(\frac{1}{1-\nu}\right) \sum_{z \in Z} p_z(1 + r_z)$. Similar to the DC plan case, substitute in the first-order condition regarding optimal savings and substitute in $\frac{\partial x^B}{\partial \varphi} = \left(\frac{1}{1-\nu}\right) \sum_{z \in Z} p_z(1 + r_z)$ to get

$$\frac{\partial \Delta V^B}{\partial \varphi} = \beta \sum_{z \in Z} p_z(1 + r_z) \sum_{z \in Z} p_z [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))]. \quad (4)$$

4.3.1 Uncertain Planning Horizon Channel

Now compare the two cases. First, consider how $\frac{\partial \Delta V^C}{\partial \varphi}$ and $\frac{\partial \Delta V^B}{\partial \varphi}$ compare in the case where there is no uncertainty regarding the rate of return.

$$\begin{aligned}\frac{\partial \Delta V^C}{\partial \varphi} &= \beta(1 - \nu)(1 + r) [u'(c_2^{C1}) - u'(c_2^{C0})] . \\ \frac{\partial \Delta V^B}{\partial \varphi} &= \beta(1 + r) [u'(c_2^{B1}) - u'(c_2^{B0})]\end{aligned}$$

The death probability $\nu \in (0, 1)$ directly reduces the impact of increasing the present value cost of the DC plan benefit on the value of retirement relative to the DB plan.

In this case, $\frac{\partial \Delta V^B}{\partial \varphi} > \frac{\partial \Delta V^C}{\partial \varphi}$ when

$$\begin{aligned}\beta(1 + r) [u'(c_2^{B1}) - u'(c_2^{B0})] &> \beta(1 - \nu)(1 + r) [u'(c_2^{C1}) - u'(c_2^{C0})] \\ \frac{1}{1 - \nu} &> \frac{u'(s^{C1}(\varphi)(1 + r)) - u'(s^{C0}(\varphi)(1 + r))}{u'(x(\varphi) + s^{B1}(\varphi)(1 + r)) - u'(x(\varphi) + s^{B0}(\varphi)(1 + r))} \equiv H(\varphi)\end{aligned} \quad (5)$$

¹¹Appendix A derives step-by-step all equations presented in this section.

Intuitively, for $\varphi = 0$ and small values of φ this inequality holds with $\nu \in (0, 1)$. The left-hand side $\frac{1}{1-\nu}$ is fixed for a given death probability, while the right-hand side is a ratio of differences in marginal utilities, which depends on how second-period consumption shifts with φ . When φ is small, both DB and DC benefits contribute little to retirement resources. On the DC side, savings increase with φ . This means a higher φ raises DC period two consumption slightly, which reduces marginal utility, but the effect is modest because φ is small. On the DB side, savings decrease with φ , but $x(\varphi)$ is also a linear function increasing in φ . When φ is small, this DB cumulative change in second-period consumption is relatively small. Thus, the numerator and denominator of the right-hand side ratio are of comparable order for small φ , so the ratio is close to 1. Since $\frac{1}{1-\nu} > 1$, the inequality is satisfied for sufficiently small φ . However, it is possible for the inequality given by (5) to fail for large φ values, because as φ grows the scaling of $u'(c_2^{C1}) - u'(c_2^{C0})$ relative to $u'(c_2^{B1}) - u'(c_2^{B0})$ differs. We discuss this result formally in Appendix A.2.1.

4.3.2 Uncertain Rate of Return Channel

Next, consider the case with $\nu = 0$ and only uncertainty regarding the rate of return. Letting $\Delta u'(c_2^i(r_z)) \equiv u'(c_2^{i1}(r_z)) - u'(c_2^{i0}(r_z))$ results in

$$\begin{aligned}\frac{\partial \Delta V^C}{\partial \varphi} &= \beta \sum_{z \in Z} p_z (1 + r_z) [u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))] = \beta \mathbb{E}[(1 + r_z) \Delta u'(c_2^C(r_z))] \\ \frac{\partial \Delta V^B}{\partial \varphi} &= \beta \sum_{z \in Z} p_z (1 + r_z) \sum_{z \in Z} p_z [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))] = \beta \mathbb{E}[1 + r_z] \mathbb{E}[\Delta u'(c_2^B(r_z))]\end{aligned}$$

Notice that $\frac{\partial \Delta V^C}{\partial \varphi}$ is the expectation of a product while $\frac{\partial \Delta V^B}{\partial \varphi}$ is the product of two expectations.¹² Therefore

$$\frac{\partial \Delta V^C}{\partial \varphi} = \beta \mathbb{E}[1 + r_z] \mathbb{E}[\Delta u'(c_2^C(r_z))] + \beta \text{Cov}(1 + r_z, \Delta u'(c_2^C(r_z))).$$

Generally, with concave utility, $\text{Cov}(1 + r_z, \Delta u'(c_2^C(r_z))) < 0$. When returns r_z are high, agents have larger period two consumption regardless of whether they worked or not in the first period, so marginal utilities fall. However, marginal utility falls more for the state with the larger savings (working in the first period), which shrinks $\Delta u'(c_2^C(r_z)) = u'(c_2^{i1}(r_z)) - u'(c_2^{i0}(r_z))$. Hence, $1 + r_z$ and $\Delta u'(c_2^C(r_z))$ tend to move in opposite directions. More uncertainty (higher variance of r_z) increases the absolute magnitude of the covariance term. Therefore, the DC derivative shrinks relative to the DB derivative as return uncertainty rises.

¹²Recall $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \text{Cov}(X, Y)$.

In this case, we have $\frac{\partial \Delta V^B}{\partial \varphi} > \frac{\partial \Delta V^C}{\partial \varphi}$ if

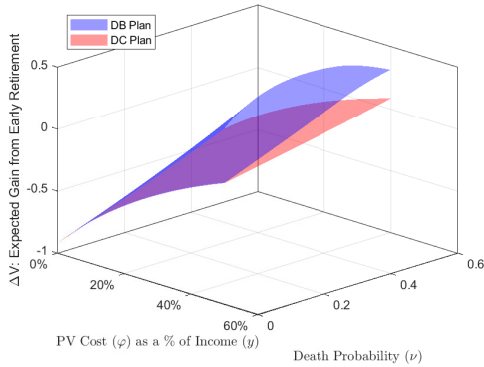
$$\begin{aligned} \beta \mathbb{E}[1 + r_z] \mathbb{E}[\Delta u'(c_2^B(r_z))] &> \beta \mathbb{E}[1 + r_z] \mathbb{E}[\Delta u'(c_2^C(r_z))] + \beta \text{Cov}(1 + r_z, \Delta u'(c_2^C(r_z))) \\ &\quad - \frac{\text{Cov}(1 + r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1 + r_z]} > \mathbb{E}[\Delta u'(c_2^C(r_z))] - \mathbb{E}[\Delta u'(c_2^B(r_z))] \equiv I(\varphi) \end{aligned} \quad (6)$$

Similar to the case discussed in subsection 4.3.1, for sufficiently small values of φ , the right-hand side of the inequality is close to zero as consumption levels in the two plans are similar for small φ . The left-hand side is positive, as $\text{Cov}(1 + r_z, \Delta u'(c_2^C(r_z)))$ is negative for $u''(c) < 0$. However, as discussed in 4.3.1, as φ increases $u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))$ falls much more sharply than $u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))$. Therefore, for sufficiently large φ values, the inequality can fail, as discussed formally in Appendix A.2.2.

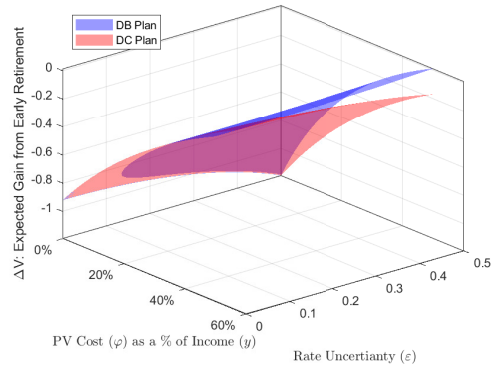
4.3.3 Numerical Illustration

This subsection plots $\Delta V^i = V_1^{i1} - V_1^{i0}$ for both plan types $i \in \{B, C\}$. We will fix parameters and consider first how ΔV^B and ΔV^C vary when there is no return risk, and there is only uncertainty regarding the timing of death.¹³ Figure 3a removes uncertainty

Figure 3: Illustrating the Uncertain Horizon and Uncertain Returns Channels



(a) Uncertain Horizon Channel



(b) Uncertain Returns Channel

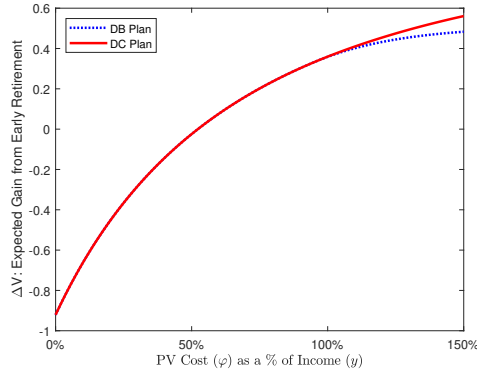
regarding investment return and considers how the expected gain from period one retirement for each plan type is affected by the death probability. This is done by holding the present value cost φ the same between plans, and varying the death probability ν . When the death probability is zero, the expected gain from period one retirement is exactly the same for both plan types. Raising the death probability raises the expected gain from retiring in period one, as agents are less likely to benefit from working and saving in period one. However, the effect of raising ν is not as strong for those in DC plans, and we see that as the death probability increases, the gap between ΔV^B and ΔV^C widens.

¹³Specifically, we set the discount factor β equal to 0.96, set initial assets a equal to 1, set income from employment y equal to 1, and set the leisure value of retirement ℓ_R equal to 1. Additionally, we assume CRRA utility $u(c) = \frac{c^{1-\theta}}{1-\theta}$ with $\theta = 2$.

Next, Figure 3b shows how ΔV^B and ΔV^C vary when the death probability is zero, so there is no uncertainty regarding agents' planning horizon, but with varying rate of return uncertainty. Specifically, let $Z \in \{L, H\}$ with $p_L = p_H = 0.5$ and $r_L = r_s - \varepsilon$ and $r_H = r_s + \varepsilon$ for $r_s = (1/\beta) - 1$. We then consider variations in ε . Notice that in the case where $\varepsilon = 0$, both plan types have the same expected gain from retiring in period one. However, increasing the rate of return uncertainty greatly reduces ΔV^C relative to ΔV^B .

Finally, Figure 4 shows that it is possible under very high plan present values for the expected gain of retiring in period one to be higher for those in the DC plan relative to those in the DB plan. When plan present value is small, the primary role of the benefit

Figure 4: Results Under High φ Values: Certain Horizon and Certain Returns



is to support retirement consumption in period two. A DB plan is more effective at this because the promised payment is guaranteed. This makes early retirement relatively more attractive under a DB when the plan size is sufficiently small. By contrast, when the plan's size is large, the DC benefit provides flexibility: workers can consume part of it in period one and still carry sufficient resources into retirement. When the plan sizes are so large that their primary use is no longer to fund period two consumption, early retirement becomes less feasible under a DB plan.

The results in this section show that when the present value cost of a plan is relatively low, uncertainty in investment returns and in the planning horizon leads individuals with DB plans to gain more from retiring early than those with DC plans. However, under sufficiently high plan present value, the ranking can reverse, with DC participants benefiting more from early retirement than their DB counterparts. In Section 3, we showed that after controlling for detailed individual and job characteristics, workers with DB plans tend to retire earlier than those with DC plans. This finding indicates that, in the U.S. context, plan present values tend to be low enough that the two uncertainty channels highlighted here make early retirement more attractive for DB plan holders.

5 Quantitative Model

5.1 Model Environment

5.1.1 Setting

Time is discrete and continues forever, with each period in the model representing one quarter. Agents who populate the model may live to terminal age $\bar{a}$ but face a death probability dependent on age (a), ν_a . Each period, a unit mass of age 30 agents enters the model while a unit mass dies at various ages. An agent could be employed, unemployed, or retired in any period. When agents enter the model, they draw a fixed income type $y \sim F(y)$. Income equals y when employed, and the unemployment benefit equals a fraction ϕ of employment income. Agents are heterogeneous in their age (a), income type (y), current labor force status, level of assets (b), accumulated DB plan tenure (γ_B), and current retirement plan type (i_p) with $i_p \in \{N \text{ (None)}, B \text{ (DB Plan)}, H \text{ (Hybrid)}, C \text{ (DC plan)}\}$.

Just as in the data, it is possible for agents in the model to move between different retirement plan types when changing jobs. Let $p(i_p|y)$ be the probability of a new job offering retirement plan type i_p given the agent's income type y . Agents with DC, hybrid, or no retirement plan will accumulate benefits through their assets (b). For individuals in a DB plan, the benefit received in retirement is a formula-based benefit $x_B(a, y, \gamma_B)$ that depends on age, income type, and accumulated DB plan tenure. We parameterize this benefit formula to match typical DB plan rules in the data; Section 6.1 describes the resulting benefit schedule in detail.¹⁴

5.1.2 Movements Between Employment States and Retirement Plans: Endogenous Retirement

Employed agents face exogenous job loss probability δ_a , while the probability of regaining employment if unemployed is η_a . Recall, when an agent enters a new job, they draw plan type $i_p \in \{N, B, H, C\}$, with probability $p(i_p|y)$. We treat retirement plan type as a characteristic of the job offer rather than an additional choice made by the worker after accepting the job. This assumption is consistent with the institutional structure of private retirement coverage around our calibration period. In 1992, only 3.0% of private employers sponsored both DB and DC plans, and this share fell to 2.6% in 1993 (U.S. General Accounting Office, 1996).¹⁵ While movements between employment and unem-

¹⁴We assume no uncertainty in DB plan payments, as the Pension Benefit Guaranty Corporation provides federal insurance for most private pensions against default risk due to plan sponsor insolvency.

¹⁵Moreover, these figures do not imply that workers could choose between the two plans. (U.S. General Accounting Office, 1996) could not determine whether the plans covered the same employees and generally classified DB plans as primary and DC plans as supplementary. Thus, even the share of employers sponsoring both plan types provides an upper bound on the prevalence of direct within-job DB-versus-DC choice.

ployment are exogenous, the choice to retire is endogenous, so that at the end of each period agents decide whether or not to retire.

5.1.3 Consumption and Savings Decisions

Agents get utility $u(c, \ell)$ from consumption c and leisure ℓ . Let ℓ_P denote leisure enjoyed if participating in the labor force, and ℓ_R denote leisure enjoyed in retirement. All individuals enter the model with zero assets and can save in one-period assets.¹⁶ However, individual investments and investments made in DC plans are subject to an uncertain rate of return r_r with $r_r = r_s + \epsilon$ and $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$.

All agents can save in financial assets whose return is uncertain. We treat these assets as a reduced-form portfolio of risky and safe financial investments, with return risk calibrated to a representative portfolio observed in the data. Those in DC plans, however, get an extra payment equal to the fraction f_c of their employment income when employed invested into the risky asset from their employer (e.g., an employer contribution to a 401(k)).¹⁷ Only those in DB and hybrid retirement plans can contribute a portion of their income in exchange for a certain outcome. Those with a DB plan receive $x_B(a, y, \gamma_B)$ each period in retirement, while those in hybrid plans have their employer contributions $f_c \times y$ invested at a safe interest rate r_s .¹⁸ In retirement, all agents receive κ_a from the government that they can access at a certain age (representing Social Security benefits).

Finally, agents receive a utility value from leaving bequests (dying with positive asset values). This assumption is common in the literature (see Gourinchas and Parker (2002), De Nardi, French and Jones (2010), Daminato and Padula (2024), among others) and will allow our model to more closely match savings behavior over the life-cycle and the observation that many individuals die with non-trivial positive levels of assets. Let $\zeta(b, y, \gamma_B)$ denote the value of dying with assets b , income y , and DB plan tenure γ_B . As many DB plans offer benefits to spouses after the death of the plan-holder, we include average expected spousal DB plan payments in the value of bequests.

¹⁶Following Kaplan and Violante (2010), we assume that entrants begin with zero assets. This level of starting assets is near the median for this age implied by the Survey of Income and Program Participation (SIPP). We additionally impose a zero borrowing limit, as in French (2005), De Nardi, French and Jones (2010), French and Jones (2011), and Heiland and Li (2012). This assumption is often used in retirement models, as the central decision is how much agents save and accumulate in preparation for retirement.

¹⁷The model treats DC-plan balances (including employer contributions) as fully liquid financial wealth. In practice, withdrawals from tax-qualified DC accounts (e.g., 401(k)s) prior to age 59.5 typically incur an additional tax penalty. We abstract from this by not distinguishing between liquid and tax-deferred assets, which keeps the state space tractable. In the calibrated model, DC participants do not begin retiring until after age 60 (with most retiring after 65) and accumulate assets up to retirement, so early withdrawals are not quantitatively important in equilibrium. If anything, incorporating an early-withdrawal penalty would further discourage retirement before age 60 under DC coverage and would likely strengthen the model-implied effects of the DB-to-DC transition.

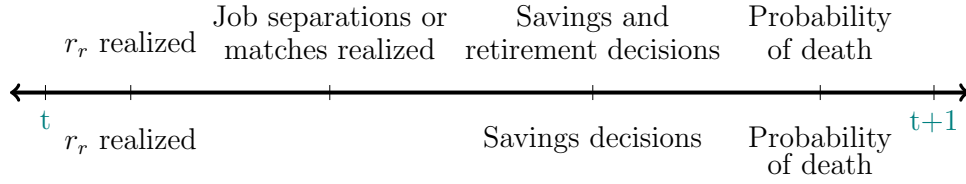
¹⁸In the model, this mimics cash-balance “hybrid” plans where individuals with these plans get access to a known amount of assets in retirement, and the amount is not subject to any uncertain rate of return or other source of risk.

5.1.4 Timing

Figure 5 summarizes the timing of events in the model. At the beginning of each period, agents observe the actual risky rate of return r_r received on the investments made last period. When making savings and retirement decisions in the current period, agents do not know the actual rate of return r_r that they will realize on their investments. After observing the realized value of r_r , an agent who is not retired could lose their job if

Figure 5: Model Timing

Not Retired:



Retired:

they entered the period employed. Similarly, agents could find a job if they entered the period unemployed. Agents then decide how much to consume and save in the current period. Still, the level of assets available in the next period is uncertain and depends on the agent's savings and the next-period realization of r_r . At this time, individuals in the labor force decide whether to remain in the labor force or retire in the next period. At the end of each period, all agents face an age-dependent death probability ν_a .¹⁹

5.2 Value Functions

Let $V_a^E(y, b, i_p, \gamma_B)$ denote the value of employment for an age a individual of income type y who has assets b , retirement plan i_p , and accumulated DB plan tenure γ_B .

$$\begin{aligned} V_a^E(y, b, i_p, \gamma_B) = & \max_{s; R \in \{0,1\}} \left\{ u(c, \ell_P) + \nu_a \beta \mathbb{E} [\zeta(b'_E(s, i_p), y, \gamma_B)] \right. \\ & + (1 - R)(1 - \nu_a) \beta [(1 - \delta_{a+1}) \mathbb{E} [V_{a+1}^E(y, b'_E(s, i_p), i_p, \gamma'_B)] + \delta_{a+1} \mathbb{E} [V_{a+1}^U(y, b'_E(s, i_p), \gamma'_B)]] \\ & \left. + R(1 - \nu_a) \beta \mathbb{E} [V_{a+1}^R(y, b'_E(s, i_p), \gamma'_B)] \right\} \end{aligned} \quad (7)$$

Agents choose how much to save s and whether to remain in the labor force or retire before the next period $R \in \{0, 1\}$. The budget constraint for the employed is as follows: consumption and savings equal the amount of assets the agent has, b , plus income y .

$$c + s = b + y \quad (8)$$

¹⁹Appendix D considers two alternative model setups where agents can re-enter the labor force after retirement and where we consider a Social Security design where Social Security income depends not only on working income but also on the age of retirement. The main results are robust to these extensions.

Agents choose how much to save s , but the amount of assets they end up with in the next period is subject to the risky rate of return r_r , which is not observed until the start of the next period. Let $b'_E(s, i_p)$ denote realized assets in the next period given savings s made when employed with a retirement plan of type i_p . In addition to the individual's personal savings, those enrolled in a DC or hybrid plan also receive an employer contribution of $f_c \times y$ to savings. The employer contribution is invested at the risky rate of return for those in a DC plan, while agents with a hybrid retirement plan have their employer contribution invested at a guaranteed safe rate of return r_s .

$$b'_E(s, i_p) = (1 + r'_r)(s + yf_c(\mathbb{1}_{i_p=C})) + (1 + r_s)yf_c(\mathbb{1}_{i_p=H}) \quad (9)$$

After making savings and retirement decisions, agents face a probability $(1 - \nu_a)$ of surviving into the next period. If the agent does not survive, they receive value $\mathbb{E}[\zeta(b'_E(s, i_p), y, \gamma_B)]$, the expected utility value from the bequests they will leave. Employed workers who survive and who did not choose to retire then become unemployed with probability δ_{a+1} and remain employed with probability $(1 - \delta_{a+1})$.

Next, let $V_a^U(y, b, \gamma_B)$ denote the value of unemployment for an age a individual of income type y who has assets b , and accumulated DB plan tenure γ_B .

$$\begin{aligned} V_a^U(y, b, \gamma_B) = & \max_{s; R \in \{0,1\}} \left\{ u(c, \ell_P) + \nu_a \beta \mathbb{E}[\zeta(b'_U(s), y, \gamma_B)] + R(1 - \nu_a) \beta \mathbb{E}[V_{a+1}^R(y, b'_U(s), \gamma_B)] \right. \\ & + (1 - R)(1 - \nu_a) \beta (1 - \eta_{a+1}) \mathbb{E}[V_{a+1}^U(y, b'_U(s), \gamma_B)] \\ & \left. + (1 - R)(1 - \nu_a) \beta \eta_{a+1} \sum_{i_p \in \{N, B, H, C\}} p(i_p | y) \mathbb{E}[V_{a+1}^E(y, b'_U(s), i_p, \gamma_B)] \right\} \quad (10) \end{aligned}$$

Agents choose how much to save s and whether to remain in the labor force or retire before the next period $R \in \{0, 1\}$. The budget constraint for the unemployed is

$$c + s = b + y\phi. \quad (11)$$

Unemployed agents receive unemployment income $y\phi$. Their consumption and savings equal their income plus their pool of existing assets. Let $b'_U(s)$ denote realized next-period assets given savings s and realized next-period risky rate of return r'_r when unemployed.

$$b'_U(s) = (1 + r'_r)s \quad (12)$$

After making savings and retirement decisions, these agents face probability $(1 - \nu_a)$ of surviving into the next period. Unemployed agents who survive and did not choose to retire then become employed in the next period with probability η_{a+1} and remain unemployed with probability $(1 - \eta_{a+1})$. When agents enter employment, they also realize

a retirement plan type associated with their new job. The probability that their new job will have retirement plan type $i_p \in \{N, B, H, C\}$ is $p(i_p|y)$.

Finally, $V_a^R(y, b, \gamma_B)$ denotes the value of retirement for an age a agent of income type y with assets b and accumulated DB plan tenure γ_B .

$$V_a^R(y, b, \gamma_B) = \max_s \{u(c, \ell_R) + \nu_a \beta \mathbb{E} [\zeta(b'_R(s), y, \gamma_B)] + (1 - \nu_a) \beta \mathbb{E} [V_{a+1}^R(y, b'_R(s), \gamma_B)]\} \quad (13)$$

In retirement, the agent receives an age-dependent benefit from the government, replacing κ_a of their employment income. Specifically, $\kappa_a = 0$ for all a values lower than the minimum age set by the government to receive this retirement benefit and $\kappa_a = \kappa$ for all ages a at and above the minimum age.²⁰ Consumption plus savings equals the government retirement income plus the agent's existing assets b . In retirement, DB tenure is fixed at the level accumulated prior to retirement, so DB income each period is $x_B(a, y, \gamma_B)$.

$$c + s = y\kappa_a + b + x_B(a, y, \gamma_B) \quad (14)$$

In retirement, if agents save s , then their realized assets in the next period given the realized next-period rate of return r'_r are as follows.

$$b'_R(s) = (1 + r'_r)s \quad (15)$$

At the end of each period, the agent faces an age-dependent death probability ν_a .

5.3 Equilibrium

Definition 1: *Recursive Equilibrium*

Given the exogenous transition probabilities $\{\nu, \delta, \eta\}$, the return process for r'_r , plan-offer probabilities $\{p(i_p | y)\}$, and the government-provided benefit schedule $\{\kappa_a\}$, a recursive equilibrium consists of:

1. Value functions $\{V_a^E(y, b, i_p, \gamma_B), V_a^U(y, b, \gamma_B), V_a^R(y, b, \gamma_B)\}$;
2. Policy functions $\{s_a^{E*}(y, b, i_p, \gamma_B), R_a^{E*}(y, b, i_p, \gamma_B)\}$, $\{s_a^{U*}(y, b, \gamma_B), R_a^{U*}(y, b, \gamma_B)\}$, and $\{s_a^{R*}(y, b, \gamma_B)\}$

²⁰We assume all retired agents receive social security income after a certain age. Although there are papers that discuss how a relatively small fraction of employees do not participate in Social Security (Morrill and Westall, 2019; Kim et al., 2024), several studies show that the impact of Social Security benefits on average retirement age is relatively small (Diamond and Hausman, 1984; Samwick, 1998; Kim et al., 2024). Furthermore, our focus is to compare optimal decisions made under DB versus DC plans. Appendix D considers a model extension where Social Security income depends not only on employment income, but also on the age of retirement. Our main results are robust to this modeling change.

3. A stationary cross-sectional distribution $\psi = (m^E, m^U, m^R)$, where $m^E(a, y, b, i_p, \gamma_B)$ is the mass of employed agents and $m^U(a, y, b, \gamma_B)$ and $m^R(a, y, b, \gamma_B)$ are the masses of unemployed and retired agents.

such that:

1. **Optimality:** For each age and possible state, (V_a^E, V_a^U, V_a^R) satisfy equations (7)–(13), and the policy functions $\{s_a^{E*}, R_a^{E*}, s_a^{U*}, R_a^{U*}, s_a^{R*}\}$ solve (7)–(13) subject to (8)–(14) and (9)–(15).
2. **Consistency of transitions:** Given the equilibrium policy functions, the distribution $\psi = (m^E, m^U, m^R)$ is generated from the policy functions, taking into account aging, death, job loss and job finding, endogenous retirement, plan draws upon job finding, and entry of new cohorts as described in the model environment.

6 Results

6.1 Calibration

We solve the model introduced in Section 5 under the following assumption regarding agents' utility. If agents are employed or unemployed, they receive utility $u(c, \ell_P) = \frac{c^{1-\theta}-1}{1-\theta} + \ell_P$, while retired agents enjoy utility $u(c, \ell_R) = \frac{c^{1-\theta}-1}{1-\theta} + \ell_R$. When comparing the utility of labor force participation and retirement, what is relevant is the difference between leisure values $\ell_R - \ell_P$. Therefore, we normalize $\ell_P = 0$ and estimate the value of ℓ_R relative to this normalization. Following De Nardi, French and Jones (2010), the value of bequests $\zeta(b, y, \gamma_B)$ is derived from agents' utility function, so that the value of leaving bequests totaling a discounted expected value of z brings bequest value $\zeta_c \frac{(z+\chi)^{1-\theta}-1}{1-\theta}$.

We calibrate the model to the 1990s before much of the transition to DC plans occurred, and will use the calibrated model to simulate the movement towards DC plans that followed. Table 4 lists directly assigned model parameters. β is the quarterly discount factor for which we choose a standard value corresponding to an annual 4% risk-free rate of return, with the quarterly risk-free rate of return $r_s = (1/\beta) - 1$. As previously discussed, we normalize $\ell_P = 0$ and will later estimate the value of ℓ_R . We normalize the median income earned in employment to 1, and take the relative 25th and 75th percentile values from the Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS).²¹ This results in $y \in \{0.6704, 1, 1.5610\}$.

²¹While we use 1992-1994 data wherever possible, the OEWS estimates for all industries on 25th, 50th, and 75th percentile annual earnings is only first available in 2001. While we rely on the earliest available 2001 estimates for the relative income values, it is worth noting that these estimates do not notably change from year to year.

Values for unemployment insurance and Social Security replacement rates are taken from estimates provided by the U.S. Department of Labor²² and the U.S. Social Security Administration, respectively. While employer contributions to DC plans can vary, we set f_c to reflect that the average employer contribution has been near 4.6% of employee income, as reported by U.S. General Accounting Office (1996) and Vanguard (2024). The report by U.S. General Accounting Office (1996) relied on data collected in 1992, aligning well with the 1992-1994 time frame of our other targets.

While DB plan benefits can vary, most follow a benefit formula where the payout in retirement is equal to a benefit multiplier times the final average salary, multiplied by the years of service. The typical benefit multiplier is 0.015 to 0.019 (U.S. Bureau of Labor Statistics, 1994*a,b*); we apply a benefit multiplier of 0.017 at the midpoint of this range. Furthermore, many plans require a minimum of seven years of service and impose a maximum of around 30 years of service after which benefits stop accruing. We follow this formula carefully, imposing a minimum of 7 years of service and a maximum of 30 years of service, after which individuals are free to continue working, but their DB benefit in retirement remains fixed (U.S. Bureau of Labor Statistics, 1994*a,b*). The quarterly age-dependent unemployment-to-employment (UE) and employment-to-unemployment (EU) transition probabilities are estimated using CPS data, displayed in Figure 6.²³ The age-dependent death probabilities in the model are taken directly from the U.S.

Table 4: Directly Assigned Parameter Values

Parameter	Description	Value/Source
β	Quarterly discount factor	0.990 (Standard, 4% annual risk-free return)
ϕ	Unemployment insurance replacement rate	0.459 (U.S. Department of Labor)
κ	Social security replacement rate	0.420 (Social Security Admin.)
f_c	Employer DC/hybrid plan contribution	0.046 (U.S. General Accounting Office, 1996)
η_a	Age-dependent UE probability	CPS estimate (Figure 6)
δ_a	Age-dependent EU probability	CPS estimate (Figure 6)
ν_a	Age-dependent death probability	Social Security Admin. Actuarial Life Table
$p(i_p y)$	Pr. of drawing each retirement plan	HRS estimates (1992–1994)
$x_B(a, y, \gamma_B)$	DB benefit formula	(U.S. Bureau of Labor Statistics, 1994 <i>a,b</i>)

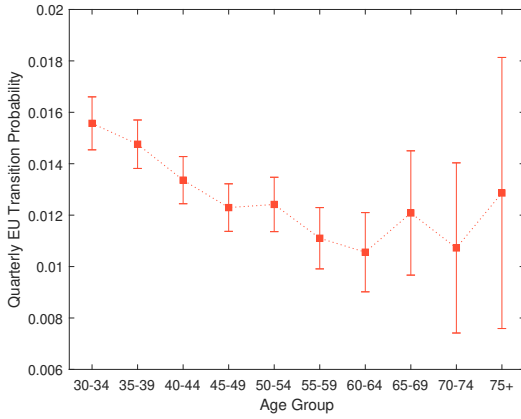
Social Security Administration’s Actuarial Life Table²⁴ and adjusted to reflect quarterly probabilities. Finally, the probabilities of drawing each of the four retirement plans when entering a job, conditional on income, are taken from the HRS data spanning 1992-1994, before the notable switch from DB towards DC plans.

²²The average unemployment insurance replacement rate is computed by the U.S. Department of Labor (DOL) (see https://oui.doleta.gov/unemploy/ui_replacement_rates.asp). We take the earliest available data from 1997, using the replacement rate computed under the DOL “Replacement Ratio 1” measure computed as the weighted average of the claimants’ weekly benefit amount divided by the product of the claimants’ normal hourly wage and 40 hours per week, where ratios greater than or equal to 2 are excluded as outliers.

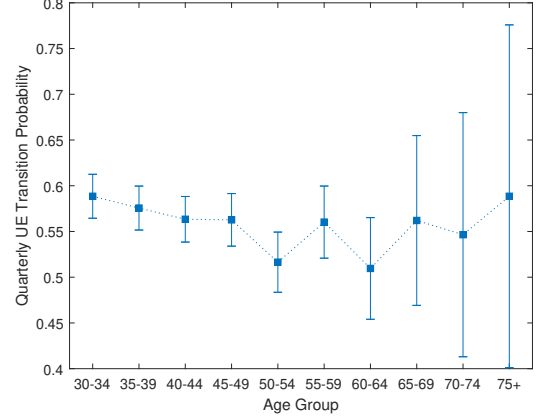
²³The CPS data used spans 1992-1999. Estimates using only 1992-1994 data do not greatly differ. The longer time horizon provides a significantly larger sample size, resulting in narrower confidence intervals.

²⁴<https://www.ssa.gov/oact/STATS/table4c6.html>

Figure 6: Estimated Quarterly Transition Probabilities by Age from CPS



(a) EU Transition Probability



(b) UE Transition Probability

After assigning the model parameters listed in Table 4, we jointly calibrate the remaining parameters to align the model with relevant features of the data. Table 5 displays the jointly calibrated parameters, their estimated values, descriptions, and a comparison of the targeted data moments and corresponding model-generated moments. In calibrating the standard deviation σ_ϵ of shocks to the risky rate of return, we target the coefficient of variation of a constructed portfolio, consisting of equity and safe financial assets. Asset weights are taken from the Employee Benefit Research Institute (EBRI) and the Investment Company Institute (ICI) 401(k) database for 1996.²⁵ The value-weighted CRSP index is used for equity and the U.S. one-month T-bill is used for safe financial assets (Fama and French, 2023).²⁶ Next, the key parameter in the model driving the incentive to retire is the retirement leisure value ℓ_R . This parameter is calibrated to target the percent of the population retired at age 65+ in 1994 from the CPS.²⁷ The bequest value parameter ζ_c is an important parameter influencing incentives to spend vs. save in retirement. A higher ζ_c value increases agents' bequest motive and results in agents consuming a smaller fraction of the savings built up before retirement. ζ_c therefore targets the percentage change in median net worth between ages 55-64 and age 75+ computed from the Survey of Consumer Finances. The agents' risk aversion θ is also an important parameter affecting savings, not only in retirement but throughout the life-cycle. The risk aversion parameter θ is chosen so that the model reasonably matches the median

²⁵The EBRI/ICI database collects annual data on participants' 401(k) plan allocations and 1996 data is the earliest available observations from the 1990s. Following their definition, equity includes equity funds, employer stock, and 60% of balanced funds. More details about the dataset can be found at [https://www.ebri.org/retirement/401\(k\)-database](https://www.ebri.org/retirement/401(k)-database).

²⁶Index data are from the May 2026 archive from the Kenneth R. French Data Library https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. The estimated coefficient of variation is calculated using data from 1992 to 2020, and it does not notably change when excluding 2020 data.

²⁷Prior to 1994, the CPS provides no information regarding whether individuals were out of the labor force due to retirement or due to another reason.

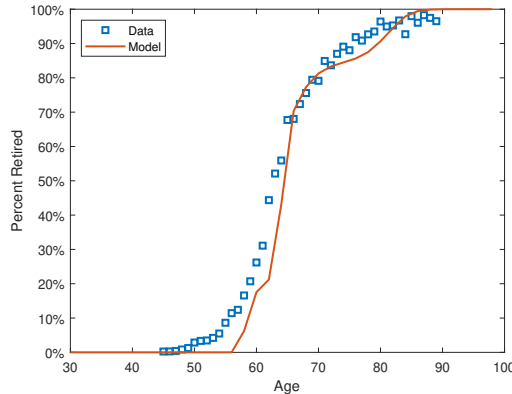
Table 5: Jointly Calibrated Parameter Values

Parameter	Estimate	Targeted Moment	Data	Model
σ_ϵ	0.0239	Quarterly returns coefficient of variation	2.66	2.70
ℓ_R	0.7574	Percent retired if age 65+	84.53%	84.54%
ζ_c	8.4437	% Change median net worth 55–64 to 75+	-23.64%	-22.86%
θ	1.1197	Age 55–64 median net worth to income ratio	4.81	4.68
χ	0.0295	Percent retired: age 70–74 to 65–69	1.17	1.16

age 55–64 net worth to income ratio.²⁸ Finally, the parameter χ affects the curvature of the bequest function, following De Nardi, French and Jones (2010). We calibrate χ by targeting the fraction of individuals who are retired between ages 70–74 relative to those between 65–69. A larger χ reduces the marginal utility cost of setting aside modest bequests, making late-life consumption and leisure relatively more attractive. In contrast, a smaller χ implies a stronger incentive to continue working in order to accumulate larger bequests. Thus, the choice of χ directly affects the model’s predictions for labor force participation at older ages. Calibrating χ to match observed retirement behavior in the 65–74 age range ensures that the model captures how bequest motives interact with late-life labor supply decisions.

In addition to the targeted moments displayed in Table 5, the model matches the percent of all individuals retired at each age reasonably well. As in the data, some early

Figure 7: Percent Retired by Age (Untargeted)



Note: The data displayed in this figure is from the CPS in 1994, as we calibrate the model to the early 1990s. Prior to 1994, there was no information available regarding whether individuals were not in the labor force due to retirement or due to another reason.

retirement occurs, with around 20% of agents retiring by age 60. The percentage of individuals retired increases drastically around the age of 65, and then levels off, with some individuals continuing to work into their 70s. Appendix D considers specifications with labor-force reentry and retirement-age-dependent Social Security benefits. Both preserve the main life-cycle patterns while generating additional retirement at earlier

²⁸For the third and fourth moments, we use data from the 1992 wave of the Survey of Consumer Finances (SCF) <https://www.federalreserve.gov/econres/scfindex.htm>.

ages, and the main counterfactual results remain robust.

Table 6: Predicted Retirement Age by Plan Type (Untargeted)

Retirement Plan	Model	Data (See Table 3)
Defined Benefit (DB)	62.04	61.44 [60.41, 62.47]
Defined Contribution (DC)	66.92	66.68 [64.90, 68.46]

While all individuals face the same death probabilities, the retirement plan type that they spend the most time in has a significant impact on when agents choose to retire. In our calibration, we chose no empirical targets relating to different behaviors between individuals enrolled in different retirement plan types. However, Table 6 shows the behavior that the model predicts aligns with the data, showing that those in DC plans tend to work longer.

6.2 Counterfactual: Movement from DB to DC Plans

Now, we use the quantitative model to evaluate the effects of the movement from DB to DC retirement plans from 1992 through 2018. We first estimate the overall effects of the transition on retirement, asset accumulation, and welfare. We then examine how allowing saving to respond earlier in the life cycle affects the retirement response before decomposing the effects into differences in plan structure and generosity, return risk, and longevity risk. Table 7 shows the change in retirement plan enrollment rates in the HRS data since the early 1990s for each earnings group. In the HRS data, the percentage enrolled in each plan type differs by income, just as in the model. While the probability of enrollment in a DC plan increased notably for all earnings groups, the percentage enrolled in a hybrid plan or no retirement plan remained relatively fixed but also fluctuated.

Table 7: Plan Enrollment Probabilities by Earnings Percentile: 1992–94 to 2016–18

	25th Percentile		50th Percentile		75th Percentile	
	1992–94	2016–18	1992–94	2016–18	1992–94	2016–18
No Plan	0.6874	0.6748	0.3400	0.2831	0.1973	0.2052
DB Plan	0.1610	0.0793	0.3874	0.0731	0.5181	0.1396
Hybrid Plan	0.0144	0.0032	0.0183	0.0125	0.0492	0.0033
DC Plan	0.1372	0.2427	0.2543	0.6313	0.2354	0.6519

First, this subsection compares three counterfactual economies, the first with the 1992–94 enrollment probabilities used in subsection 6.1, the second with the new retirement plan enrollment probabilities observed in 2016–18 (see Table 7), and the third with no plan and hybrid plan enrollment rates fixed to their 1992–94 values while the shift out of DB plan enrollment is accounted for by an increase in DC plan enrollment (see Table

8). The purpose of considering this last counterfactual economy is to isolate the effects of the shift out of DB plans alone, without the smaller fluctuations in no plan and hybrid plan enrollment occurring simultaneously.²⁹

Table 9 displays the result of the shifting retirement plan enrollment rates. First, notice that the effects of considering all plan enrollment changes that occurred versus only considering the transition out of DB towards DC plans are very similar. The percentage of agents retired at ages ranging from 65-70+ is lower in the steady state of the economy with higher DC plan enrollment. The effect is greatest when looking at earlier retirement

Table 9: Estimated Effects of Retirement Plan Transition

	All Plan Enrollment Changes 1992–2019	Only transition out of DB
% Retired at Age 65–69	-1.5pp	-1.3pp
% Retired at Age 65	-14.3pp	-13.7pp
% Retired at Age 70+	-0.3pp	-0.3pp
Average Age 55 Assets	+26.3%	+25.4%
Average Age 80 Assets	+31.2%	+30.4%
Average Age 95 Assets	-4.0%	-4.0%
Entrant Consumption Equivalent Variation	-1.6%	-1.8%

by age 65. This relatively large drop in the percentage of individuals retired at age 65 is consistent with the CPS data shown in Figure 2.

Next, average assets at age 55 (when agents are primarily still working) and at age 80 (when the majority of agents are retired), are higher after the transitions away from DB plans. The increase in assets before retirement reflects one of the two insurance responses emphasized in the paper. In addition to working longer, households respond to greater reliance on DC plans by accumulating more assets before retirement. However, average assets for those who reach age 95 are lower, as individuals with greater reliance on DC plans consume a larger portion of their savings if they reach later retirement.

The last row of Table 9 reports the entrant consumption equivalent variation (CEV)

²⁹Specifically, in the third counterfactual economy where no plan and hybrid plan enrollment rates are fixed to their 1992–94 values and the shift out of DB plan enrollment is accounted for by an increase in DC plan enrollment the plan enrollment rates are as in the 2016–18 columns of Table 8.

Table 8: Plan Enrollment Probabilities: Shift Out of DB Plans Only

	25th Percentile		50th Percentile		75th Percentile	
	1992–94	2016–18	1992–94	2016–18	1992–94	2016–18
No Plan	0.6874	0.6874	0.3400	0.3400	0.1973	0.1973
DB Plan	0.1610	0.0793	0.3874	0.0731	0.5181	0.1396
Hybrid Plan	0.0144	0.0144	0.0183	0.0183	0.0492	0.0492
DC Plan	0.1372	0.2189	0.2543	0.5686	0.2354	0.6139

Notice in this case, the no plan and hybrid plan enrollment probabilities are unchanged from their 1992–94 values, the DB plan enrollment probabilities reflect the estimated 2016–18 values, and the DC plan enrollment probabilities are then set so that total enrollment probabilities sum to one.

associated with the change in retirement-plan enrollment rates. The CEV converts the change in the entrant’s full expected lifetime utility, including utility from consumption, labor-force and retirement leisure, and bequests, into a uniform percentage adjustment to consumption in the baseline economy that would make the entrant indifferent between the baseline and counterfactual economies. The estimated CEV of -1.6% implies that the full set of changes in plan enrollment between 1992-1994 and 2016-2018 reduces welfare by an amount equivalent to permanently reducing baseline consumption by 1.6% at every age and contingency. When we isolate only the shift out of DB plans and into DC plans, the corresponding welfare loss is 1.8% of baseline consumption. These losses arise despite the increase in accumulated assets because the movement away from DB plans changes the structure and present value of retirement benefits and places greater return and longevity risk on individuals. The decomposition that follows separates the contributions of these plan-design and uncertainty channels.³⁰

6.2.1 The Role of Additional Saving as Insurance

An important difference between our counterfactual studying the effects of the transition from DB to DC plans and Bairoliya (2019) is the role of saving before age 55. In Bairoliya (2019), a subset of DB participants is reassigned to DC coverage at age 55 while their assets, Social Security wealth, and wage process are held fixed. In our model, the transition toward DC plans affects decisions earlier in the life cycle, allowing households to adjust their saving before reaching retirement ages. This distinction is important because saving more and working longer provide two ways for households to respond to the additional return and longevity risk associated with DC plans. When households accumulate more assets earlier in life, they have less need to respond to these risks by working longer at older ages.

Table 10 shows the importance of this saving response. To make the results directly comparable with Bairoliya (2019), the table reports the percentage change in participation among ages 65-69 rather than the percentage-point change. In the full model, the DB-to-DC transition increases participation among ages 65-69 by 4.40% . We then consider a counterfactual that removes the effects of the transition on saving before age 55. In this case, participation increases by 5.81% , compared with 5.37% in Bairoliya (2019).

Allowing saving to adjust over the life cycle reduces the predicted participation response at older ages. Households exposed to greater risk under DC plans partly insure themselves by accumulating additional assets before age 55, reducing the need to insure against the same risks by working longer. When this earlier saving response is removed, our estimated participation effect becomes notably larger and is close to the estimate in

³⁰In Appendix D we show that the results presented in this section are robust to alternative model specifications that allow agents to re-enter the workforce and adjust Social Security income based on retirement age.

Table 10: Percentage Change in 65–69 Participation

Full Model	No savings effects before 55	Bairoliya (2019)
4.40%	5.81%	5.37%

Notes: The “No saving effects before 55” counterfactual follows the approach in Bairoliya (2019) by exogenously reassigning a subset of age 55 DB participants to DC plans. Existing DB pension wealth is converted into DC wealth. Other assets and pre-55 states are held fixed. See Bairoliya (2019), Table 7.

Bairoliya (2019). This comparison highlights that saving and retirement decisions operate jointly as households respond to the additional risks associated with DC plans.

6.2.2 Decomposition

To better understand why the shift from DB to DC plans reduces retirement at earlier ages and affects total expected utility, we decompose the full-model effects into components attributable to (i) differences in plan structure and generosity and (ii) the additional risks faced by DC participants. We do so by computing the full counterfactual change in the baseline economy with both risk channels present (investment return risk and longevity risk), and then re-running the same counterfactual in auxiliary economies that “shut down” one or both channels. This gives us an additive decomposition where the full-model change is the sum of a plan-design/present value component (the change when both risk channels are removed), a component driven by uncertain returns (return risk), a component driven by uncertain horizon (longevity risk), and a residual interaction term that captures non-additives from both uncertainty channels operating simultaneously.³¹

Table 11 shows that the decline in retirement at 65 is driven primarily by the uncertain returns channel. In Panel A, the full-model effect on retirement by 65 is a 13.7 percentage point reduction, and the majority of this effect is accounted for by the total uncertainty component (9.8 pp), with a smaller effect from differences in plan structure and generosity alone (3.9 pp). Panel B indicates that this total uncertainty effect is mostly attributable to uncertain investment returns. With a certain planning horizon, uncertain returns reduce retirement by 65 by 14.9 pp, while uncertain horizon risk on its own has a small effect (+2.3 pp). The interaction term (+2.8 pp) indicates that when both risks are present, the combined effect is less negative than the sum of the separate effects.

The decomposition clarifies the mechanisms behind savings and welfare responses. Panel A shows that the welfare effects are split across both sources, with roughly two-thirds of the change in welfare due to plan-structure/present-value differences alone and one-third due to the added uncertainty faced. Within the uncertainty component, Panel B shows that uncertain horizon risk generates -0.5% of the -0.6% CEV from the uncertainty components alone, with a near-zero contribution from uncertain returns and a small interaction term (-0.1%). Thus, although return risk is the primary driver of delayed

³¹The construction follows the nested-counterfactual formulas described in the notes to Table 11.

Table 11: Decomposition of DB-to-DC Transition Effects into Plan Design and Uncertainty Channels

Panel A. Full effect split into plan structure and generosity vs. total uncertainty			
Outcome	Full model	plan structure and generosity	Total uncertainty
% Retired at 65 (pp)	-13.7	-3.9	-9.8
% Retired at 70+ (pp)	-0.3	-1.5	+1.2
Avg. age-55 assets (%)	+25.4	+20.6	+4.8
Entrant CEV (%)	-1.8	-1.2	-0.6
Panel B. Total uncertainty split into returns, horizon, and interaction			
Outcome	Uncertain returns	Uncertain horizon	Returns×Horizon (interaction)
% Retired at 65 (pp)	-14.9	+2.3	+2.8
% Retired at 70+ (pp)	+0.7	+3.1	-2.6
Avg. age-55 assets (%)	-7.9	+15.4	-2.7
Entrant CEV (%)	+0.0	-0.5	-0.1

Notes: The “Full model” column reports the effects of the DB-to-DC transition where both return risk and longevity risk are present (see column “Only transition out of DB” in Table 9). The “Plan structure and generosity” column reports the corresponding counterfactual when both uncertainty channels are shut down. The “Total uncertainty” column shows the difference between the Full model and Plan structure and generosity columns. Panel B defines the channel contributions using the nested counterfactuals: Uncertain returns = (Uncertain returns only)– (Certain); Uncertain horizon = (Uncertain horizon only)– (Certain); Interaction = Full– (Uncertain returns only)– (Uncertain horizon only)+ (Certain).

retirement at age 65, uncertain-horizon risk is the primary driver of the welfare loss attributable to uncertainty. Households respond to the additional risks associated with DC plans through both asset accumulation and continued work, but these responses do not eliminate the welfare costs. The next section considers whether allowing households to annuitize assets can make saving a more effective source of retirement insurance.

6.3 Counterfactual: Effects of Option to Convert Assets to Actuarially Fair Annuity Upon Retirement

Next, we analyze the effects of a policy that permits agents to convert any portion of their assets (b) upon retirement into an actuarially fair annuity. In the period of retirement, agents choose how much of their accumulated assets to trade for a single-life annuity at an actuarially fair price. To purchase an annuity with quarterly payment p_N , an agent entering retirement at age a will have to give up assets $b_N(a, p_N)$, totaling the expected discounted annuity payments over their remaining lifetime.

$$\begin{aligned}
 b_N(a, p_N) &= p_N + (1 - \nu_a)\beta p_N + (1 - \nu_a)(1 - \nu_{a+1})\beta^2 p_N + \dots \\
 b_N(a, p_N) &= p_N + p_N \sum_{k=1}^{\bar{a}-a} \beta^k \prod_{j=0}^{k-1} (1 - \nu_{a+j})
 \end{aligned} \tag{16}$$

Let $V_a^{R1}(y, b, \gamma_B)$ denote the value of first entering retirement at age a with assets b and accumulated DB plan tenure γ_B . When initially entering retirement, agents now choose p_N , their fixed quarterly annuity payment. To receive p_N , the agent gives up $b_N(a, p_N)$ in assets.

$$V_a^{R1}(y, b, \gamma_B) = \max_{p_N \geq 0} \{V_a^R(y, b - b_N(a, p_N), \gamma_B, p_N)\} \quad (17)$$

Once retired with chosen payment p_N , the retired agent's value function is almost identical to that in the baseline economy (13), except that p_N now becomes a state variable. Notice that p_N does not enter the expected value of bequests, as we assume the policy provides the option to purchase single-life annuities that are terminated upon the agent's death.

$$\begin{aligned} V_a^R(y, b, \gamma_B, p_N) = \max_s \{ & u(c, \ell_R) + \nu_a \beta \mathbb{E} [\zeta(b'_R(s), y, \gamma_B)] \\ & + (1 - \nu_a) \beta \mathbb{E} [V_{a+1}^R(y, b'_R(s), \gamma_B, p_N)] \} \end{aligned} \quad (18)$$

Similarly, the retired agent's budget constraint differs from (14) only in that the budget now includes the quarterly payment p_N .

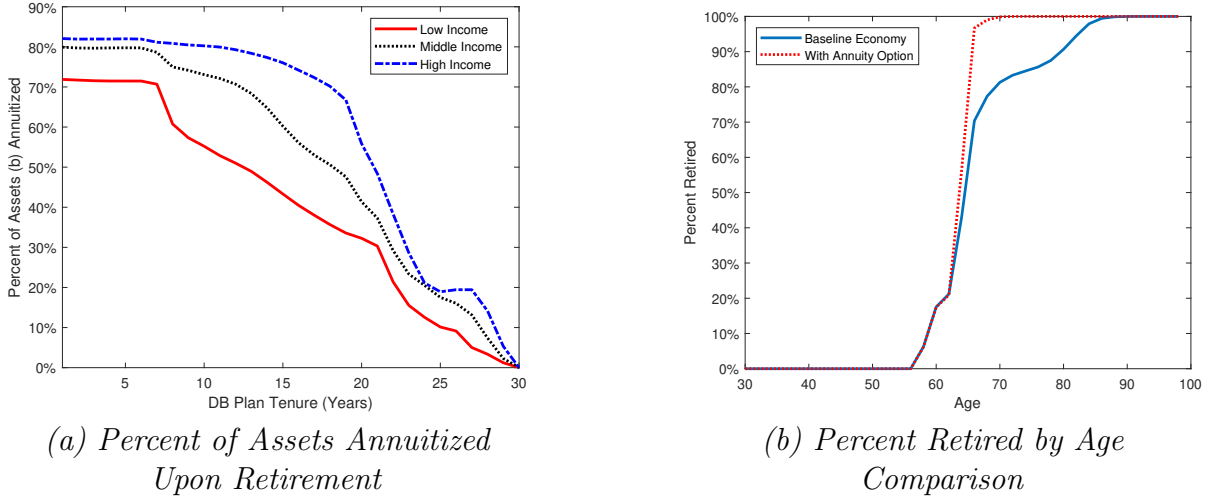
$$c + s = y\kappa_a + p_N + b + x_B(a, y, \gamma_B) \quad (19)$$

The values of employment and unemployment, previously given by (7) and (10) respectively, now include a transition to V_{a+1}^{R1} rather than V_{a+1}^R upon initial retirement. We write this change to V_a^E in the following equation, and the change to V_a^U follows analogously.

$$\begin{aligned} V_a^E(y, b, i_p, \gamma_B) = \max_{s, R \in \{0,1\}} \{ & u(c, \ell_P) + \nu_a \beta \mathbb{E} [\zeta(b'_E(s, i_p), y, \gamma_B)] \\ & + (1 - R)(1 - \nu_a) \beta [(1 - \delta_{a+1}) \mathbb{E} [V_{a+1}^E(y, b'_E(s, i_p), i_p, \gamma'_B)] + \delta_{a+1} \mathbb{E} [V_{a+1}^U(y, b'_E(s, i_p), \gamma'_B)]] \\ & + R(1 - \nu_a) \beta \mathbb{E} [V_{a+1}^{R1}(y, b'_E(s, i_p), \gamma'_B)] \} \end{aligned} \quad (20)$$

Figure 8 displays the effects of the fair annuity option policy in the baseline economy, where enrollment in each retirement plan type matches 1992-94 enrollment probabilities. Figure 8a shows the average percent of assets (b) that agents annuitize when entering retirement, conditional on DB plan tenure and income type. Those who have high DB tenure, and so already receive a high annuity-type payment in retirement, annuitize a smaller portion of assets. This figure also shows that lower-income agents annuitize a smaller portion of wealth relative to high-income agents, and there is no group that chooses to annuitize all of their assets. Notice that annuitized wealth does not enter the expected value of leaving bequests $\mathbb{E} [\zeta(b'_E(s, i_p), y, \gamma_B)]$ because the payments p_N are assumed to be solely for a single-life annuity. Agents do not annuitize all of their wealth because doing so would mean leaving no bequests, and with the curvature in

Figure 8: Effects of Annuity Policy in Baseline (1992-94) Economy



the bequest function specified and calibrated in section 6.1, this would weigh heavily on their total expected discounted continuation value. The motive to leave some assets as bequests results in lower wealth agents, who are more likely to be low-income, annuitizing a smaller fraction of their assets. Figure 8b shows the effect of the fair annuity policy on retirement timing, with a much larger mass of agents choosing to retire at 65 relative to the baseline economy, as the fair annuity option reduces the motive to work longer as a precaution against uncertainty.

Table 12 displays the aggregate effects of introducing the actuarially fair annuity option in the baseline economy, where retirement plan enrollment probabilities match those in 1992-94, and the effects of introducing the annuity option after the transition into DC plans. Introducing the policy in the baseline economy generates an entrant consumption equivalent variation of +0.3% and increases retirement by age 65.³² Average assets at age 55 are higher, as the value of asset accumulation is greater when there is an option to annuitize. By age 80, agents on average hold fewer non-annuitized assets, as they choose to annuitize a portion of their wealth. However, those who live to 95 have higher non-annuitized wealth, as they rely on consuming this wealth less in retirement.

Table 12: Effects of Offering Actuarially Fair Annuity Option Upon Retirement

	Baseline Economy (1992-94 Enrollment)	After DC Transition (2016-18 Enrollment)
% Retired at Age 65	+23.4pp	+28.7pp
% Retired at Age 70+	+10.7pp	+11.0pp
Average Age 55 (Non-Annuitized) Assets	+17.2%	+15.5%
Average Age 80 (Non-Annuitized) Assets	-27.9%	-43.1%
Average Age 95 (Non-Annuitized) Assets	+11.3%	+20.7%
Entrant Consumption Equivalent Variation	+0.3%	+0.6%

³²The 0.3% CEV implies that access to the annuity option raises welfare by an amount equivalent to increasing the entrant's consumption by 0.3% throughout the baseline allocation.

The last column of Table 12 displays the effects of the annuity option after the transition away from DB plans. The benefit of the annuity option is greater in this case, when fewer agents get annuity-like benefits from a DB retirement plan. Introducing the annuity option in the 2016-2018 economy generates an entrant CEV of +0.6% and has a notable impact on encouraging earlier retirement. Similar to introducing the annuity option in the baseline economy, we observe that agents initially save more on average by age 55, when they can benefit more from asset accumulation with the option to annuitize those assets. Non-annuitized assets are lower on average at age 80, but those who live to age 95 rely less on non-annuitized wealth to fund retirement consumption and have greater amounts of non-annuitized assets on average after the policy is introduced.

Overall, the annuity option reduces exposure to both return and longevity risk after retirement. Agents consequently accumulate more assets before retirement, retire earlier, and draw down their remaining non-annuitized wealth more slowly at advanced ages. The welfare and retirement effects are larger after the transition toward DC plans, when fewer individuals receive annuity-like income through DB plans.

7 Conclusion

The transition from DB to DC plans changes both the risks households face and the flexibility they have over retirement resources. DB plans provide guaranteed lifetime benefits that largely insure against return and longevity risk, while DC plans provide earlier access to retirement wealth but expose individuals to greater uncertainty. Households can respond to these additional risks by saving more and by working longer. We show that these responses operate jointly, with additional saving before retirement reducing the need to respond to retirement risk through continued work at older ages.

We then decompose the effects of the DB-to-DC transition into differences in plan structure and generosity, return risk, and longevity risk. The delayed-retirement response is driven primarily by greater exposure to return risk, as working longer provides insurance against poor investment realizations. In contrast, longevity risk is the primary source of the welfare loss attributable to uncertainty, while differences in plan structure and generosity also account for a substantial share of the overall welfare loss.

Finally, we consider whether accumulated assets can provide better retirement insurance when retirees can freely convert any fraction of their assets into actuarially fair annuities. Annuitization reduces exposure to longevity and return risk after retirement, raises welfare, and reduces the need to work longer as insurance against retirement risk. The gains are larger after the transition toward DC plans, when fewer households receive guaranteed lifetime income through DB plans.

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Supplemental Appendix

A Additional Illustrative Model Results

This appendix derives step-by-step the results stated in Section 4. First, A.1 solves for the savings Euler equation and shows formally that optimal savings is strictly increasing in DC plan present value and strictly decreasing in DB plan present value. Next, A.2 derives equations (3) and (4) in the main text and formally shows that the conditions given by (5) and (6) hold under standard CRRA utility for modest levels of plan present value φ , but can fail for sufficiently high φ .

Recall the period one and expected period two values.

$$V_1^i(a) = \max_{R \in \{0,1\}, s} \left\{ u((1-R)y + f^i + a - s) + R\ell_R + \beta(1-\nu)\mathbb{E}[V_2^i(s)] \right\}$$

$$\mathbb{E}[V_2^i(s)] = \ell_R + \sum_{z \in Z} p_z u(x^i + s(1+r_z))$$

Therefore, given the choice of R , the agents solve the following optimal savings problem.

$$V_1^i(a) = \max_s \left\{ u((1-R)y + f^i + a - s) + R\ell_R + \beta(1-\nu)\ell_R + \beta(1-\nu) \sum_{z \in Z} p_z u(x^i + s(1+r_z)) \right\}$$

A.1 Optimal Savings

The first-order condition for optimal savings is

$$\frac{\partial V_1^i(a)}{\partial s} = -u'((1-R)y + f^i + a - s) + \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z)u'(x^i + s(1+r_z)) = 0.$$

This gives us the Euler equation in the main text

$$u'((1-R)y + f^i + a - s) = \beta(1-\nu) \sum_{z \in Z} [p_z(1+r_z)u'(x^i + s(1+r_z))] .$$

Proposition 1: $\frac{\partial s}{\partial f^i} > 0$ and $\frac{\partial s}{\partial x^i} < 0$ if $u''(c) < 0$.

Proof. Define $G(s, f^i, x^i)$ as the first-order condition residual

$$G(s, f^i, x^i) \equiv u'((1-R)y + f^i + a - s) - \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z)u'(x^i + s(1+r_z))$$

where an interior optimum s satisfies $G(s, f^i, x^i) = 0$. Now compute the partial derivatives of $G(s, f^i, x^i)$, using $u'(c) > 0$ and $u''(c) < 0$.

$$\frac{\partial G(s, f^i, x^i)}{\partial s^i} = -u''((1-R)y + f^i + a - s) - \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z)^2 u''(x^i + s(1+r_z))$$

$$\frac{\partial G(s, f^i, x^i)}{\partial f^i} = u''((1-R)y + f^i + a - s)$$

$$\frac{\partial G(s, f^i, x^i)}{\partial x^i} = -\beta(1-\nu) \sum_{z \in Z} p_z(1+r_z) u''(x^i + s(1+r_z))$$

Recall the standard concave utility assumption $u'(c) > 0$ and $u''(c) < 0$ we made in the main text, and apply the implicit function theorem. The implicit function theorem tells us that

$$\begin{aligned} \frac{\partial s}{\partial f^i} &= - \left(\frac{\partial G(s, f^i, x^i)}{\partial s^i} \right)^{-1} \left(\frac{\partial G(s, f^i, x^i)}{\partial f^i} \right) \\ \frac{\partial s}{\partial x^i} &= - \left(\frac{\partial G(s, f^i, x^i)}{\partial s^i} \right)^{-1} \left(\frac{\partial G(s, f^i, x^i)}{\partial x^i} \right). \end{aligned}$$

Therefore

$$\frac{\partial s}{\partial f^i} = \frac{-u''((1-R)y + f^i + a - s)}{-u''((1-R)y + f^i + a - s) - \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z)^2 u''(x^i + s(1+r_z))}$$

With $u''(c) < 0$, both numerator and denominator are positive. $u''((1-R)y + f^i + a - s) < 0$ so that $-u''((1-R)y + f^i + a - s) > 0$ and $u''(x^i + s(1+r_z)) < 0$ so that $-u''((1-R)y + f^i + a - s) - \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z)^2 u''(x^i + s(1+r_z)) > 0$. Therefore

$$\frac{\partial s}{\partial f^i} > 0 \quad \text{if} \quad u''(c) < 0.$$

Similarly,

$$\frac{\partial s}{\partial x^i} = \frac{-1 \times \left[-\beta(1-\nu) \sum_{z \in Z} p_z(1+r_z) u''(x^i + s(1+r_z)) \right]}{-u''((1-R)y + f^i + a - s) - \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z)^2 u''(x^i + s(1+r_z))}$$

With $u''(c) < 0$, the denominator is positive, as in $\frac{\partial s}{\partial f^i}$. Then, since $u''(x^i + s(1+r_z)) < 0$ the numerator $-1 \times \left[-\beta(1-\nu) \sum_{z \in Z} p_z(1+r_z) u''(x^i + s(1+r_z)) \right]$ is less than zero. Thus

$$\frac{\partial s}{\partial x^i} < 0 \quad \text{if} \quad u''(c) < 0.$$

□

A.2 Optimal Retirement

As in the main text, define

$$\begin{aligned}\Delta V^i \equiv V_1^{i1}(a) - V_1^{i0}(a) &= \ell_R + u(f^i + a - s^{i1}) - u(y + f^i + a - s^{i0}) \\ &\quad + \beta(1 - \nu) \sum_{z \in Z} p_z [u(x^i + s^{i1}(1 + r_z)) - u(x^i + s^{i0}(1 + r_z))].\end{aligned}$$

First, consider an agent with a DC plan ($i = C$) where $f^C \geq 0$ and $x^C = 0$. We will consider how ΔV^C changes when the present value cost of providing the DC plan ($\varphi = f^C$) increases.

$$\begin{aligned}\frac{\partial \Delta V^C}{\partial \varphi} &= u'(c_1^{C1}) \left(1 - \frac{\partial s^{C1}}{\partial \varphi}\right) - u'(c_1^{C0}) \left(1 - \frac{\partial s^{C0}}{\partial \varphi}\right) \\ &\quad + \beta(1 - \nu) \sum_{z \in Z} p_z \left[u'(c_2^{C1}(r_z))(1 + r_z) \left(\frac{\partial s^{C1}}{\partial \varphi}\right) - u'(c_2^{C0}(r_z))(1 + r_z) \left(\frac{\partial s^{C0}}{\partial \varphi}\right) \right]\end{aligned}$$

Now substitute in the savings first-order condition where $i = C$ for $R \in \{0, 1\}$.

$$\begin{aligned}u'(c_1^{C1}) &= \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C1}(r_z)) \\ u'(c_1^{C0}) &= \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C0}(r_z))\end{aligned}$$

$$\begin{aligned}\frac{\partial \Delta V^C}{\partial \varphi} &= \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C1}(r_z)) \left(1 - \frac{\partial s^{C1}}{\partial \varphi}\right) - \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C0}(r_z)) \left(1 - \frac{\partial s^{C0}}{\partial \varphi}\right) \\ &\quad + \beta(1 - \nu) \sum_{z \in Z} p_z \left[u'(c_2^{C1}(r_z))(1 + r_z) \left(\frac{\partial s^{C1}}{\partial \varphi}\right) - u'(c_2^{C0}(r_z))(1 + r_z) \left(\frac{\partial s^{C0}}{\partial \varphi}\right) \right] \\ \frac{\partial \Delta V^C}{\partial \varphi} &= \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C1}(r_z)) \left(1 - \frac{\partial s^{C1}}{\partial \varphi}\right) - \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C0}(r_z)) \left(1 - \frac{\partial s^{C0}}{\partial \varphi}\right) \\ &\quad + \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C1}(r_z)) \left(\frac{\partial s^{C1}}{\partial \varphi}\right) - \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C0}(r_z)) \left(\frac{\partial s^{C0}}{\partial \varphi}\right)\end{aligned}$$

Notice that the terms multiplied by $\left(\frac{\partial s^{C0}}{\partial \varphi}\right)$ and $\left(\frac{\partial s^{C1}}{\partial \varphi}\right)$ cancel, and we are left with

$$\begin{aligned}\frac{\partial \Delta V^C}{\partial \varphi} &= \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C1}(r_z)) - \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u'(c_2^{C0}(r_z)) \\ \frac{\partial \Delta V^C}{\partial \varphi} &= \beta(1 - \nu) \sum_{z \in Z} p_z (1 + r_z) [u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))].\end{aligned}$$

Next, consider an agent with a DB plan where $f^B = 0$ and $x^B \geq 0$. To be consistent with the DC plan case, we will consider a change in the present value cost of providing the DB plan

φ . Recall $\varphi = \frac{x^B(1-\nu)}{\sum_{z \in Z} p_z(1+r_z)}$, so that $x^B = \varphi \left(\frac{1}{1-\nu} \right) \sum_{z \in Z} p_z(1+r_z)$.

$$\begin{aligned} \frac{\partial \Delta V^B}{\partial \varphi} &= \left(\frac{\partial x}{\partial \varphi} \right) \left[u'(c_1^{B1}) \left(-\frac{\partial s^{B1}}{\partial x^B} \right) - u'(c_1^{B0}) \left(-\frac{\partial s^{B0}}{\partial x^B} \right) \right] \\ &+ \beta(1-\nu) \left(\frac{\partial x^B}{\partial \varphi} \right) \sum_{z \in Z} p_z \left[u'(c_2^{B1}(r_z)) \left[1 + (1+r_z) \frac{\partial s^{B1}}{\partial x^B} \right] - u'(c_2^{B0}(r_z)) \left[1 + (1+r_z) \frac{\partial s^{B0}}{\partial x^B} \right] \right] \end{aligned}$$

Now substitute in the savings first order condition conditional on this plan type.

$$\begin{aligned} u'(c_1^{B1}) &= \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z) u'(c_2^{B1}(r_z)) \\ u'(c_1^{B0}) &= \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z) u'(c_2^{B0}(r_z)) \end{aligned}$$

This results in

$$\begin{aligned} \frac{\partial \Delta V^B}{\partial \varphi} &= \left(\frac{\partial x^B}{\partial \varphi} \right) \left[\beta(1-\nu) \sum_{z \in Z} p_z(1+r_z) u'(c_2^{B1}(r_z)) \left(-\frac{\partial s^{B1}}{\partial x^B} \right) - \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z) u'(c_2^{B0}(r_z)) \left(-\frac{\partial s^{B0}}{\partial x^B} \right) \right] \\ &+ \beta(1-\nu) \left(\frac{\partial x^B}{\partial \varphi} \right) \sum_{z \in Z} p_z \left[u'(c_2^{B1}(r_z)) \left[1 + (1+r_z) \frac{\partial s^{B1}}{\partial x^B} \right] - u'(c_2^{B0}(r_z)) \left[1 + (1+r_z) \frac{\partial s^{B0}}{\partial x^B} \right] \right] \end{aligned}$$

$$\begin{aligned} \frac{\partial \Delta V^B}{\partial \varphi} &= \beta(1-\nu) \left(\frac{\partial x^B}{\partial \varphi} \right) \sum_{z \in Z} p_z \left[u'(c_2^{B1}(r_z))(1+r_z) \left(-\frac{\partial s^{B1}}{\partial x^B} \right) - u'(c_2^{B0}(r_z))(1+r_z) \left(-\frac{\partial s^{B0}}{\partial x^B} \right) \right] \\ &+ \beta(1-\nu) \left(\frac{\partial x^B}{\partial \varphi} \right) \sum_{z \in Z} p_z \left[u'(c_2^{B1}(r_z)) \left[1 + (1+r_z) \frac{\partial s^{B1}}{\partial x^B} \right] - u'(c_2^{B0}(r_z)) \left[1 + (1+r_z) \frac{\partial s^{B0}}{\partial x^B} \right] \right] \end{aligned}$$

Notice that terms multiplied by $(1+r_z) \left(\frac{\partial s^{B1}}{\partial x^B} \right)$ and $(1+r_z) \left(\frac{\partial s^{B0}}{\partial x^B} \right)$ cancel out, resulting in

$$\frac{\partial \Delta V^B}{\partial \varphi} = \beta(1-\nu) \left(\frac{\partial x^B}{\partial \varphi} \right) \sum_{z \in Z} p_z [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))]$$

Finally, because $x^B = \varphi \left(\frac{1}{1-\nu} \right) \sum_{z \in Z} p_z(1+r_z)$ for present value cost of providing the plan φ , we can substitute in $\frac{\partial x^B}{\partial \varphi} = \left(\frac{1}{1-\nu} \right) \sum_{z \in Z} p_z(1+r_z)$.

$$\begin{aligned} \frac{\partial \Delta V^B}{\partial \varphi} &= \beta(1-\nu) \left(\frac{1}{1-\nu} \right) \sum_{z \in Z} p_z(1+r_z) \sum_{z \in Z} p_z [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))] \\ \frac{\partial \Delta V^B}{\partial \varphi} &= \beta \sum_{z \in Z} p_z(1+r_z) \sum_{z \in Z} p_z [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))] \end{aligned}$$

A.2.1 Uncertain Planning Horizon Channel

Consider how $\frac{\partial \Delta V^C}{\partial \varphi}$ and $\frac{\partial \Delta V^{DB}}{\partial \varphi}$ compare in the case where there is no uncertainty regarding the rate of return. For this case, assume that there is only one possible rate of return r .

Specifically $Z = \{1\}$, $p_1 = 1$, and $r_1 = r$. In this case, $\frac{\partial \Delta V^C}{\partial \varphi}$ and $\frac{\partial \Delta V^B}{\partial \varphi}$ simplify to

$$\begin{aligned}\frac{\partial \Delta V^C}{\partial \varphi} &= \beta(1 - \nu)(1 + r) [u'(c_2^{C1}) - u'(c_2^{C0})] . \\ \frac{\partial \Delta V^B}{\partial \varphi} &= \beta(1 + r) [u'(c_2^{B1}) - u'(c_2^{B0})]\end{aligned}$$

This shows us that the uncertain life-expectancy, with death probability $\nu \in (0, 1)$ directly reduces the impact of increasing the present value cost of the DC plan benefit on the value of retirement.

In this case, $\frac{\partial \Delta V^B}{\partial \varphi} > \frac{\partial \Delta V^C}{\partial \varphi}$ when

$$\begin{aligned}\beta(1 + r) [u'(c_2^{B1}) - u'(c_2^{B0})] &> \beta(1 - \nu)(1 + r) [u'(c_2^{C1}) - u'(c_2^{C0})] \\ \frac{1}{1 - \nu} &> \frac{u'(s^{C1}(\varphi)(1 + r)) - u'(s^{C0}(\varphi)(1 + r))}{u'(x(\varphi) + s^{B1}(\varphi)(1 + r)) - u'(x(\varphi) + s^{B0}(\varphi)(1 + r))} \equiv H(\varphi).\end{aligned}$$

Proposition 2: *Under the assumption of constant relative risk averse (CRRA) utility, $u(c) = \frac{c^{1-\theta}}{1-\theta}$ for $\theta > 0$, there exists a φ_H^* such that $H(\varphi) < \frac{1}{1-\nu}$ if $\varphi < \varphi_H^*$ and $H(\varphi) > \frac{1}{1-\nu}$ if $\varphi > \varphi_H^*$.*

Proof. Apply the intermediate value theorem: if $H(0) < \frac{1}{1-\nu}$ and $\lim_{\varphi \rightarrow \infty} H(\varphi) > \frac{1}{1-\nu}$ then there is at least one crossing φ_H^* where $H(\varphi_H^*) = \frac{1}{1-\nu}$. We will show i) $H(\varphi)$ is a continuous function in φ so that the intermediate value theorem applies, ii) $H(0) < \frac{1}{1-\nu}$, and iii) $\lim_{\varphi \rightarrow \infty} H(\varphi) > \frac{1}{1-\nu}$.

1. First, we show $H(\varphi)$ is a continuous function in φ so that the intermediate value theorem applies. Recall our assumption of CRRA utility, so the utility function is twice continuously differentiable ($u \in C^2$), $u'(c) > 0$, and $u''(c) < 0$.

Note that each $s^{iR}(\varphi)$ is continuously differentiable in φ by the Implicit Function Theorem. In the proof of Proposition 1 we derived the explicit derivative formulas, showing the derivatives exist and are continuous given $u \in C^2$. Therefore, each mapping $\varphi \mapsto s^{C1}(\varphi), s^{C0}(\varphi), s^{B1}(\varphi), s^{B0}(\varphi)$ is continuous. Then, because $u'(c)$ and $x(\varphi)$ are continuous (recall $x(\varphi)$ is a linear function of φ), the four mappings $\varphi \mapsto u'(s^{C1}(\varphi)(1 + r)), u'(s^{C0}(\varphi)(1 + r)), u'(x(\varphi) + s^{B1}(\varphi)(1 + r)), u'(x(\varphi) + s^{B0}(\varphi)(1 + r))$ are continuous. Therefore, the numerator and denominator of $H(\varphi)$ are both continuous functions of φ .

Furthermore, the denominator of $H(\varphi)$ is nonzero. Strict concavity and the fact that first period income differs across R imply that optimal savings under $R = 1$ and $R = 0$ differ, resulting in the two marginal utilities to differ. Hence, the quotient of two continuous functions where the denominator is nonzero yields a continuous function $H(\varphi)$.

2. Now we show $H(0) < \frac{1}{1-\nu}$ for $\nu \in (0, 1)$. At $\varphi = 0$ both DB and DC plan payments are zero ($f^C = 0$ and $x^B = 0$). Therefore, agents with either plan type have savings Euler equation

$$u'((1 - R)y + a - s) = \beta(1 - \nu) [(1 + r)u'(s(1 + r))].$$

Therefore, with $\varphi = 0$, the optimal savings choices coincide across plan types in each retirement state. That is, for $\varphi = 0$

$$s^{B0}(0) = s^{C0}(0) \quad \text{and} \quad s^{B1}(0) = s^{C1}(0).$$

Consequently, the two marginal-utility differences in the definition of $H(\varphi)$ are equal

$$u'(s^{C1}(0)(1+r)) - u'(s^{C0}(0)(1+r)) = u'(x^B(0) + s^{B1}(0)(1+r)) - u'(x^B(0) + s^{B0}(0)(1+r)).$$

Thus, for $\nu \in (0, 1)$,

$$H(0) = 1 < \frac{1}{1-\nu}.$$

3. Finally, we show $\lim_{\varphi \rightarrow \infty} H(\varphi) > \frac{1}{1-\nu}$.

First, consider the savings Euler equation where there is no uncertainty regarding the rate of return, so $r_z = r$ for all z , and when utility is CRRA.

$$\begin{aligned} u'((1-R)y + f^i + a - s^{iR}) &= \beta(1-\nu)(1+r)u'(x^i + s^{iR}(1+r)). \\ ((1-R)y + f^i + a - s^{iR})^{-\theta} &= \beta(1-\nu)(1+r)(x^i + s^{iR}(1+r))^{-\theta}. \\ (1-R)y + f^i + a - s^{iR} &= (\beta(1-\nu)(1+r))^{\frac{-1}{\theta}} (x^i + s^{iR}(1+r)). \end{aligned}$$

Let $K \equiv [\beta(1-\nu)(1+r)]^{\frac{1}{\theta}} > 0$.

$$\begin{aligned} ((1-R)y + f^i + a - s^{iR}) K &= x^i + s^{iR}(1+r). \\ ((1-R)y + f^i + a) K - x^i &= s^{iR}(1+r) + K s^{iR}. \\ s^{iR} &= \frac{((1-R)y + f^i + a) K - x^i}{1+r+K}. \end{aligned}$$

Recall

$$H(\varphi) = \frac{u'(s^{C1}(\varphi)(1+r)) - u'(s^{C0}(\varphi)(1+r))}{u'(x^B(\varphi) + s^{B1}(\varphi)(1+r)) - u'(x^B(\varphi) + s^{B0}(\varphi)(1+r))}$$

Under CRRA utility this is

$$H(\varphi) = \frac{(s^{C1}(\varphi)(1+r))^{-\theta} - (s^{C0}(\varphi)(1+r))^{-\theta}}{(x^B(\varphi) + s^{B1}(\varphi)(1+r))^{-\theta} - (x^B(\varphi) + s^{B0}(\varphi)(1+r))^{-\theta}}$$

Now substitute in $x^B = \varphi^{\frac{1+r}{1-\nu}}$, $f^C = \varphi$ and the optimal savings rules s^{iR} that we solved

for

$$\begin{aligned}
H(\varphi) &= \frac{\left(\frac{(\varphi+a)K(1+r)}{1+r+K}\right)^{-\theta} - \left(\frac{(y+\varphi+a)K(1+r)}{1+r+K}\right)^{-\theta}}{\left(\varphi^{\frac{1+r}{1-\nu}} + \frac{(aK-\varphi^{\frac{1+r}{1-\nu}})}{1+r+K}(1+r)\right)^{-\theta} - \left(\varphi^{\frac{1+r}{1-\nu}} + \frac{((y+a)K-\varphi^{\frac{1+r}{1-\nu}})}{1+r+K}(1+r)\right)^{-\theta}} \\
H(\varphi) &= \frac{\left(\frac{K(1+r)}{1+r+K}\right)^{-\theta} \left[(a+\varphi)^{-\theta} - (y+\varphi+a)^{-\theta}\right]}{\left(\frac{K(1+r)}{1+r+K}\right)^{-\theta} \left[\left(\varphi^{\frac{1}{1-\nu}} + a\right)^{-\theta} - \left(\varphi^{\frac{1}{1-\nu}} + (y+a)\right)^{-\theta}\right]} \\
H(\varphi) &= \frac{(a+\varphi)^{-\theta} - (y+a+\varphi)^{-\theta}}{\left(a+\varphi^{\frac{1}{1-\nu}}\right)^{-\theta} - \left(y+a+\varphi^{\frac{1}{1-\nu}}\right)^{-\theta}}
\end{aligned}$$

First, factor out $\varphi^{-\theta}$ in the numerator

$$(a+\varphi)^{-\theta} - (y+a+\varphi)^{-\theta} = \varphi^{-\theta} \left[\left(\frac{a}{\varphi} + 1\right)^{-\theta} - \left(\frac{y+a}{\varphi} + 1\right)^{-\theta} \right]$$

Similarly, factor out $\frac{\varphi}{1-\nu}$ from the denominator

$$\begin{aligned}
&\left(a+\varphi^{\frac{1}{1-\nu}}\right)^{-\theta} - \left(y+a+\varphi^{\frac{1}{1-\nu}}\right)^{-\theta} \\
&= \left(\frac{\varphi}{1-\nu}\right)^{-\theta} \left[\left(\frac{a(1-\nu)}{\varphi} + 1\right)^{-\theta} - \left(\frac{(y+a)(1-\nu)}{\varphi} + 1\right)^{-\theta} \right]
\end{aligned}$$

Let $z \equiv \frac{\varphi}{1-\nu}$ so $\varphi = z(1-\nu)$ and consider the limit when $z \rightarrow \infty \Rightarrow \varphi \rightarrow \infty$. Substitute $\varphi = z(1-\nu)$ and write $H(\varphi)$ in terms of z .

$$\begin{aligned}
H(\varphi) &= \frac{\varphi^{-\theta} \left[\left(\frac{a}{\varphi} + 1\right)^{-\theta} - \left(\frac{y+a}{\varphi} + 1\right)^{-\theta} \right]}{\left(\frac{\varphi}{1-\nu}\right)^{-\theta} \left[\left(\frac{a(1-\nu)}{\varphi} + 1\right)^{-\theta} - \left(\frac{(y+a)(1-\nu)}{\varphi} + 1\right)^{-\theta} \right]} \\
H(z) &= \frac{(z(1-\nu))^{-\theta} \left[\left(\frac{a}{z(1-\nu)} + 1\right)^{-\theta} - \left(\frac{y+a}{z(1-\nu)} + 1\right)^{-\theta} \right]}{(z)^{-\theta} \left[\left(\frac{a}{z} + 1\right)^{-\theta} - \left(\frac{y+a}{z} + 1\right)^{-\theta} \right]} \\
H(z) &= (1-\nu)^{-\theta} \frac{\left[\left(\frac{a}{z(1-\nu)} + 1\right)^{-\theta} - \left(\frac{y+a}{z(1-\nu)} + 1\right)^{-\theta} \right]}{\left[\left(\frac{a}{z} + 1\right)^{-\theta} - \left(\frac{y+a}{z} + 1\right)^{-\theta} \right]}
\end{aligned}$$

Now take the limit as $z \rightarrow \infty$, noting that $(1 + \frac{\text{constant}}{z})^{-\theta} \rightarrow 1$, so that

$$\lim_{z \rightarrow \infty} \frac{\left[\left(\frac{a}{z(1-\nu)} + 1\right)^{-\theta} - \left(\frac{y+a}{z(1-\nu)} + 1\right)^{-\theta} \right]}{\left[\left(\frac{a}{z} + 1\right)^{-\theta} - \left(\frac{y+a}{z} + 1\right)^{-\theta} \right]} \rightarrow \frac{y}{y(1-\nu)} = \frac{1}{1-\nu}$$

Thus

$$\lim_{\varphi \rightarrow \infty} H(\varphi) = (1 - \nu)^{-\theta} (1 - \nu)^{-1} = (1 - \nu)^{-(\theta+1)}.$$

Therefore

$$\lim_{\varphi \rightarrow \infty} H(\varphi) = (1 - \nu)^{-(\theta+1)} = \frac{1}{(1 - \nu)^{\theta+1}} > \frac{1}{1 - \nu}.$$

This holds when $\nu \in (0, 1)$ and $\theta > 0$.

□

Proposition 3: φ_H^* is unique.

Proof. From the proof of Proposition 2, we showed that

$$H(\varphi) = \frac{(a + \varphi)^{-\theta} - (y + a + \varphi)^{-\theta}}{\left(a + \varphi \frac{1}{1-\nu}\right)^{-\theta} - \left(y + a + \varphi \frac{1}{1-\nu}\right)^{-\theta}}$$

under the assumption that $u(c) = \frac{u^{1-\theta}}{1-\theta}$ and defining $K \equiv [\beta(1 - \nu)(1 + r)]^{\frac{1}{\theta}} > 0$. Now we show that $H(\varphi)$ is strictly increasing in φ .

Apply the Fundamental Theorem of Calculus, which gives us $(a + \varphi)^{-\theta} - (y + a + \varphi)^{-\theta} = \theta \int_{a+\varphi}^{y+a+\varphi} t^{-\theta-1} dt$ and $(a + \varphi \frac{1}{1-\nu})^{-\theta} - (y + a + \varphi \frac{1}{1-\nu})^{-\theta} = \theta \int_{a+\varphi \frac{1}{1-\nu}}^{y+a+\varphi \frac{1}{1-\nu}} t^{-\theta-1} dt$. Therefore, the function $H(\varphi)$ can be written as

$$H(\varphi) = \frac{\int_{a+\varphi}^{y+a+\varphi} t^{-\theta-1} dt}{\int_{a+\varphi \frac{1}{1-\nu}}^{y+a+\varphi \frac{1}{1-\nu}} t^{-\theta-1} dt} \equiv \frac{H_N(\varphi)}{H_D(\varphi)}.$$

Differentiate the numerator $H_N(\varphi)$ and denominator $H_D(\varphi)$ using Leibniz rule for differentiation under the integral sign. $H'_N(\varphi) = (y + a + \varphi)^{-\theta-1} - (a + \varphi)^{-\theta-1}$ and $H'_D(\varphi) = \frac{1}{1-\nu} \left[(y + a + \varphi \frac{1}{1-\nu})^{-\theta-1} - (a + \varphi \frac{1}{1-\nu})^{-\theta-1} \right]$. Using the quotient rule:

$$H'(\varphi) = \frac{H_D(\varphi)H'_N(\varphi) - H_N(\varphi)H'_D(\varphi)}{(H_D(\varphi))^2}$$

Both $H_D(\varphi) > 0$ and $H_N(\varphi) > 0$, so to show $H'(\varphi) > 0$ we need to show

$$\begin{aligned} H_D(\varphi)H'_N(\varphi) - H_N(\varphi)H'_D(\varphi) &> 0 \\ \frac{H'_N(\varphi)}{H_N(\varphi)} &> \frac{H'_D(\varphi)}{H_D(\varphi)} \end{aligned}$$

Notice that $\frac{H'_N(\varphi)}{H_N(\varphi)}$ is the slope of $t^{-\theta}$ over an interval of length y divided by the average $t^{-\theta-1}$ over the same interval. Because $t^{-\theta}$ is convex, this ratio decreases as the starting point $a + \varphi$ increases. $\frac{H'_D(\varphi)}{H_D(\varphi)}$ is the same ratio, except with starting point $a + \varphi \frac{1}{1-\nu}$. Since $a + \varphi \frac{1}{1-\nu} > a + \varphi$, $\frac{H'_N(\varphi)}{H_N(\varphi)} > \frac{H'_D(\varphi)}{H_D(\varphi)}$. Therefore $H'(\varphi) > 0$, so $H(\varphi)$ is strictly increasing, making φ_H^* unique.

To formally show $\frac{H'_N(\varphi)}{H_N(\varphi)} > \frac{H'_D(\varphi)}{H_D(\varphi)}$ define $A(u) = \int_u^{u+y} t^{-\theta-1} dt$ and $B(u) = u^{-\theta-1} - (u+y)^{-\theta-1}$. Letting $u_1 = a + \varphi$ and $u_2 = a + \varphi \frac{1}{1-\nu}$, the inequality $\frac{H'_N(\varphi)}{H_N(\varphi)} > \frac{H'_D(\varphi)}{H_D(\varphi)}$ becomes $\frac{A(u_2)}{B(u_2)}(1-\nu) < \frac{A(u_1)}{B(u_1)}$. Since $u_2 > u_1$ and $\nu \in (0, 1)$, this inequality holds if we show $\frac{A(u)}{B(u)}$ is decreasing in u .

To show $\frac{A(u)}{B(u)}$ is decreasing in u , define $g(t) = t^{-\theta-1}$ so that $A(u) = \int_0^y g(u+s)ds$, $B(u) = g(u) - g(u+y) > 0$ and $A'(u) = -B(u)$. Then, since $A'(u) = -B(u)$,

$$\frac{\partial \frac{A(u)}{B(u)}}{\partial u} = \frac{B(u)A'(u) - A(u)B'(u)}{B(u)^2} = \frac{-B(u)^2 - A(u)(g'(u) - g'(u+y))}{B(u)^2} \equiv \frac{\mathcal{N}(u)}{B(u)^2}$$

Note that $B(u) > 0$ so the denominator is positive, so $\frac{\partial \frac{A(u)}{B(u)}}{\partial u} < 0$ if $\mathcal{N}(u) < 0$. To more easily evaluate the numerator, rewrite $\mathcal{N}(u) = \frac{1}{2} \int_0^y \int_0^y (g'(u+s) - g'(u+t))(g(u+s) - g(u+t)) ds dt$. Then for any s, t where $s \neq t$, we have that the terms $(g'(u+s) - g'(u+t))$ and $(g(u+s) - g(u+t))$ will have opposite signs (because g' is strictly increasing while g is strictly decreasing), making their product negative. Thus, we have $\frac{\partial \frac{A(u)}{B(u)}}{\partial u} < 0$ so that $\frac{H'_N(\varphi)}{H_N(\varphi)} > \frac{H'_D(\varphi)}{H_D(\varphi)}$, resulting in $H(\varphi)$ being strictly increasing in φ . $\square$

A.2.2 Uncertain Return Channel

Next, consider the case with $\nu = 0$ and only uncertainty regarding the rate of return. Letting $\Delta u'(c_2^i(r_z)) \equiv u'(c_2^{i1}) - u'(c_2^{i0})$ results in

$$\begin{aligned} \frac{\partial \Delta V^B}{\partial \varphi} &= \beta \sum_{z \in Z} p_z(1+r_z) \sum_{z \in Z} p_z [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))] \\ &= \beta \mathbb{E}[1+r_z] \mathbb{E}[\Delta u'(c_2^B(r_z))] \\ \frac{\partial \Delta V^C}{\partial \varphi} &= \beta \sum_{z \in Z} p_z(1+r_z) [u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))] \\ &= \beta \mathbb{E}[(1+r_z) \Delta u'(c_2^C(r_z))] \\ &= \beta \mathbb{E}[1+r_z] \mathbb{E}[\Delta u'(c_2^C(r_z))] + \beta \text{Cov}(1+r_z, \Delta u'(c_2^C(r_z))) \end{aligned}$$

In this case, we have $\frac{\partial \Delta V^B}{\partial \varphi} > \frac{\partial \Delta V^C}{\partial \varphi}$ if

$$\frac{\beta \mathbb{E}[1+r_z] \mathbb{E}[\Delta u'(c_2^B(r_z))] > \beta \mathbb{E}[1+r_z] \mathbb{E}[\Delta u'(c_2^C(r_z))] + \beta \text{Cov}(1+r_z, \Delta u'(c_2^C(r_z))) - \text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]} > \mathbb{E}[u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))] - \mathbb{E}[u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))] \equiv I(f)$$

Proposition 4: Under the assumption of CRRA utility, there exists an φ_I^* with $I(\varphi) < \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ if $\varphi < \varphi_I^*$ and $I(\varphi) > \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ if $\varphi > \varphi_I^*$.

Proof. Apply the intermediate value theorem: if $I(0) < \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ and $\lim_{\varphi \rightarrow \infty} I(\varphi) > \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ then there is at least one crossing φ_I^* where $I(\varphi_I^*) = \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$. We will show i) $I(\varphi)$ and $\frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ are continuous functions in φ so that the

intermediate value theorem applies, ii) $I(0) < \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$, and iii) $\lim_{\varphi \rightarrow \infty} I(\varphi) > \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$.

1. First, we show $I(\varphi)$ and $\frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ are continuous functions in φ so that the intermediate value theorem applies. Recall our assumption of CRRA utility $u(c) = \frac{c^{1-\theta}}{1-\theta}$ for $\theta > 1$ implies that the utility function is twice continuously differentiable and concave ($u \in C^2$), $u'(c) > 0$, and $u''(c) < 0$.

We start by showing $I(\varphi)$ is continuous in φ using similar arguments as the proof of Proposition 2. Note again that each $s^{iR}(\varphi)$ is continuously differentiable in φ by the Implicit Function Theorem. Therefore, each mapping $\varphi \mapsto s^{C1}(\varphi), s^{C0}(\varphi), s^{B1}(\varphi), s^{B0}(\varphi)$ is continuous. Then, because $u'(c)$ and $x(\varphi)$ are continuous (recall $x(\varphi)$ is a linear function of φ), the four mappings $\varphi \mapsto u'(s^{C1}(\varphi)(1+r_z)), u'(s^{C0}(\varphi)(1+r_z)), u'(x(\varphi) + s^{B1}(\varphi)(1+r_z)), u'(x(\varphi) + s^{B0}(\varphi)(1+r_z))$ are continuous. Therefore, the function $I(\varphi)$, which is a linear combination of these terms, is also a continuous function of φ .

Next, we show $\frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ is a continuous function of φ . Recall $\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z))) = \mathbb{E}[(1+r_z)\Delta u'(c_2^C(r_z))] - \mathbb{E}[1+r_z]\mathbb{E}[\Delta u'(c_2^C(r_z))]$. Each expectation is a finite sum of continuous terms (because $(1+r_z)$ is a fixed scalar for each z and we showed $\Delta u'(c_2^C(r_z))$ is continuous in φ). Finally, $\mathbb{E}[1+r_z]$ is a constant that is independent of φ , so dividing by it preserves continuity. Therefore, the ratio $\frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$ is continuous in φ .

2. Here we show that $I(0) < \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$. At $\varphi = 0$ both DB and DC plan payments are zero. Therefore, agents with either plan type have savings Euler equation

$$u'((1-R)y + a - s) = \beta(1-\nu) [(1+r)u'(s(1+r))].$$

Therefore, with $\varphi = 0$, the optimal savings choices coincide across plan types in each retirement state. That is, for $\varphi = 0$

$$s^{B0}(0) = s^{C0}(0) \quad \text{and} \quad s^{B1}(0) = s^{C1}(0).$$

Consequently, $\mathbb{E}[u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))] - \mathbb{E}[u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))] = 0$ so that $I(0) = 0$. Therefore, it is sufficient to show that $I(0) = 0 < \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$, or $0 > \text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))$.

We will show that $\Delta u'(c_2^C(r_z))$ is decreasing in r_z , so that it is negatively correlated with $1+r_z$. Here $\Delta u'(c_2^C(r_z)) = u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z))$ where $c_2^{CR}(r_z) = s^{CR}(\varphi)(1+r_z)$. Now differentiate $u'(c_2^C(r_z))$ with respect to r_z .

$$\frac{\delta \Delta u'(c_2^C(r_z))}{\delta r_z} = s^{C1}(\varphi)u''(c_2^{C1}(r_z)) - s^{C0}(\varphi)u''(c_2^{C0}(r_z)) \quad (21)$$

Then $\frac{\delta \Delta u'(c_2^C(r_z))}{\delta r_z} < 0$ when $s^{C1}(\varphi)u''(c_2^{C1}(r_z)) - s^{C0}(\varphi)u''(c_2^{C0}(r_z)) < 0$, which is when $s^{C1}(\varphi)u''(c_2^{C1}(r_z)) < s^{C0}(\varphi)u''(c_2^{C0}(r_z))$.

With CRRA utility and $\theta > 1$ this is

$$\begin{aligned} s^{C1}(\varphi)(-\theta)(s_2^{C1}(\varphi)(1+r_z))^{-\theta-1} &< s^{C0}(\varphi)(-\theta)(s_2^{C0}(\varphi)(1+r_z))^{-\theta-1} \\ s^{C1}(\varphi)(s^{C1}(\varphi))^{-\theta-1} &> s^{C0}(\varphi)(s^{C0}(\varphi))^{-\theta-1} \\ \frac{1}{s^{C1}(\varphi)^\theta} &> \frac{1}{s^{C0}(\varphi)^\theta} \end{aligned}$$

Which holds when

$$s^{C0}(\varphi) > s^{C1}(\varphi).$$

Recall the savings Euler equation, assuming CRRA utility and in this case $\nu = 0$.

$$\begin{aligned} (y + \varphi + a - s^{C0}(\varphi))^{-\theta} &= \beta \sum_{z \in Z} p_z(1+r_z) (s^{C0}(\varphi)(1+r_z))^{-\theta} \\ (\varphi + a - s^{C1}(\varphi))^{-\theta} &= \beta \sum_{z \in Z} p_z(1+r_z) (s^{C1}(\varphi)(1+r_z))^{-\theta} \end{aligned}$$

It follows that the optimal savings of those who work in the first period $s^{C0}(\varphi)$ is strictly greater than the optimal savings of those who retire in the first period $s^{C1}(\varphi)$ for $y > 1$ in order for the period one marginal utility to equalize to the expected discounted period two marginal utility.

3. Finally, we show $\lim_{\varphi \rightarrow \infty} I(\varphi) > \lim_{\varphi \rightarrow \infty} \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$. Recall

$$I(\varphi) = \mathbb{E} [\Delta u'(c_2^C(r_z))] - \mathbb{E} [\Delta u'(c_2^B(r_z))]$$

Furthermore, $-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z))) = \mathbb{E}[1+r_z]\mathbb{E}[\Delta u'(c_2^C(r_z))] - \mathbb{E}[(1+r_z)\Delta u'(c_2^C(r_z))]$, so that

$$\frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]} = \mathbb{E}[\Delta u'(c_2^C(r_z))] - \frac{\mathbb{E}[(1+r_z)\Delta u'(c_2^C(r_z))]}{\mathbb{E}[1+r_z]}. \quad (22)$$

So to show $\lim_{\varphi \rightarrow \infty} I(\varphi) > \lim_{\varphi \rightarrow \infty} \frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]}$, we need to show

$$\lim_{\varphi \rightarrow \infty} \mathbb{E} [\Delta u'(c_2^C(r_z))] - \mathbb{E} [\Delta u'(c_2^B(r_z))] > \lim_{\varphi \rightarrow \infty} \mathbb{E}[\Delta u'(c_2^C(r_z))] - \frac{\mathbb{E}[(1+r_z)\Delta u'(c_2^C(r_z))]}{\mathbb{E}[1+r_z]}$$

$$\lim_{\varphi \rightarrow \infty} \mathbb{E}[1+r_z]\mathbb{E} [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))] < \lim_{\varphi \rightarrow \infty} \mathbb{E} [(1+r_z) (u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z)))]$$

Where $\Delta u'(c_2^i(r_z)) = u'(c_2^{i1}(r_z)) - u'(c_2^{i0}(r_z))$. Substituting in $c_2^{B1}(r_z) = x^B(\varphi) + s^{B1}(\varphi)(1+r_z)$, $c_2^{B0}(r_z) = x^B(\varphi) + s^{B0}(\varphi)(1+r_z)$, $c_2^{C1}(r_z) = s^{C1}(\varphi)(1+r_z)$, $c_2^{C0}(r_z) = s^{C0}(\varphi)(1+r_z)$,

$x^B(\varphi) = \varphi \mathbb{E}[1 + r_z]$, and assuming CRRA utility ($u(c) = \frac{c^{1-\theta}}{1-\theta}$) the relevant inequality is

$$\begin{aligned} \lim_{\varphi \rightarrow \infty} \mathbb{E}[1 + r_z] \mathbb{E} \left[(\varphi \mathbb{E}[1 + r_z] + s^{B1}(\varphi)(1 + r_z))^{-\theta} - (\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z))^{-\theta} \right] \\ < \lim_{\varphi \rightarrow \infty} \mathbb{E} \left[(1 + r_z) \left((s^{C1}(\varphi)(1 + r_z))^{-\theta} - (s^{C0}(\varphi)(1 + r_z))^{-\theta} \right) \right] \end{aligned}$$

We will show that this inequality holds by proceeding in two steps: 1) we will show that the left-hand side involving B-terms approaches zero as $\varphi \rightarrow 0$, and 2) we will show that the right-hand side involving C-terms approaches a value strictly greater than zero.

To show $\lim_{\varphi \rightarrow \infty} \mathbb{E}[1 + r_z] \mathbb{E} \left[(\varphi \mathbb{E}[1 + r_z] + s^{B1}(\varphi)(1 + r_z))^{-\theta} - (\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z))^{-\theta} \right] = 0$. Recall the derivatives of optimal savings with respect to φ derived in the proof of Proposition 1. For those in a DB plan ($i = B$) making retirement choice $R \in \{0, 1\}$

$$\frac{\partial s^{BR}}{\partial \varphi} = \frac{\mathbb{E}[1 + r_z] \beta (1 - \nu) \sum_{z \in Z} p_z (1 + r_z) u''(\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z))}{-u''(y(1 - R) + a - s^{B0}(\varphi)) - \beta (1 - \nu) \sum_{z \in Z} p_z (1 + r_z)^2 u''(\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z))}.$$

Under CRRA utility, $u''(c) = -\theta c^{-\theta-1}$. The numerator contains $u''(\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z)) = -\theta(\varphi \mathbb{E}[1 + r_z] + O(1))^{-\theta-1}$. Hence the numerator is $O(\varphi^{-\theta-1})$. The denominator has two terms. The first term $-u''(y(1 - R) + a - s^{B0}(\varphi))$ is positive and bounded away from zero because $y(1 - R) + a - s^{B0}(\varphi)$ approaches $y(1 - R) + a$ as we showed $s^{B0}(\varphi)$ is strictly decreasing in φ . The second denominator term involves the same term as the numerator $u''(\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z)) = -\theta(\varphi \mathbb{E}[1 + r_z] + O(1))^{-\theta-1}$, which is $O(\varphi^{-\theta-1})$. Therefore $\frac{\partial s^{BR}}{\partial \varphi} = O(\varphi^{-\theta-1})$ as $\varphi \rightarrow \infty$. This means that the rate at which $s^{BR}(\varphi)$ changes becomes vanishingly small, and the difference between the two savings choices $s^{B1}(\varphi) - s^{B0}(\varphi)$ must also approach zero as $\varphi \rightarrow \infty$ because both savings functions are monotone and their slopes shrink like $\varphi^{-\theta-1}$. Now consider

$$(\varphi \mathbb{E}[1 + r_z] + s^{B1}(\varphi)(1 + r_z))^{-\theta} - (\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z))^{-\theta}$$

As φ grows $\varphi \mathbb{E}[1 + r_z]$ grows without bound, and the difference between the two terms approaches zero as $s^{B1}(\varphi) - s^{B0}(\varphi)$ approaches zero. Therefore,

$$\lim_{\varphi \rightarrow \infty} \mathbb{E}[1 + r_z] \mathbb{E} \left[(\varphi \mathbb{E}[1 + r_z] + s^{B1}(\varphi)(1 + r_z))^{-\theta} - (\varphi \mathbb{E}[1 + r_z] + s^{B0}(\varphi)(1 + r_z))^{-\theta} \right] = 0.$$

The second and final step is now to show

$$\lim_{\varphi \rightarrow \infty} \mathbb{E} \left[(1 + r_z) \left((s^{C1}(\varphi)(1 + r_z))^{-\theta} - (s^{C0}(\varphi)(1 + r_z))^{-\theta} \right) \right] > 0.$$

Recall optimal DC plan ($i = C$) savings behavior given retirement choice $R \in \{0, 1\}$:

$$u'((1 - R)y + \varphi + a - s^{CR}) = \beta(1 - \nu) \sum_{z \in Z} [p_z (1 + r_z) u'(s^{CR}(1 + r_z))]$$

so that with CRRA utility savings when working in the first period $s^{C0}(\varphi)$ are greater than savings when retiring in the first period $s^{C1}(\varphi)$. Furthermore, we found in the proof

of Proposition 1 that

$$\frac{\partial s^{CR}(\varphi)}{\partial \varphi} = \frac{-u''(y(1-R) + \varphi + a - s^{C0}(\varphi))}{-u''(y + \varphi + a - s^{C0}(\varphi)) - \beta(1-\nu) \sum_{z \in Z} p_z(1+r_z)^2 u''(s^{C0}(\varphi)(1+r_z))}$$

so $s^{C0}(\varphi)$ and $s^{C1}(\varphi)$ do not explode with φ and $\frac{\partial s^{CR}(\varphi)}{\partial \varphi} \rightarrow 0$ as $\varphi \rightarrow \infty$. Given boundedness of $s^{C0}(\varphi)$ and $s^{C1}(\varphi)$, and the ordering $s^{C0}(\varphi) > s^{C1}(\varphi)$, the difference $((s^{C1}(\varphi)(1+r_z))^{-\theta} - (s^{C0}(\varphi)(1+r_z))^{-\theta})$ is positive and bounded away from zero. Multiplying and taking the expectation preserves positivity so that

$$\lim_{\varphi \rightarrow \infty} \mathbb{E} [(1+r_z) ((s^{C1}(\varphi)(1+r_z))^{-\theta} - (s^{C0}(\varphi)(1+r_z))^{-\theta})] > 0.$$

□

Proposition 5: φ_I^* is unique.

Proof. Here we show that $\Phi(\varphi) \equiv I(\varphi) - \left(\frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]} \right)$ has the property $\Phi'(\varphi) > 0$.

Recall

$$I(\varphi) = \mathbb{E} [\Delta u'(c_2^C(r_z))] - \mathbb{E} [\Delta u'(c_2^B(r_z))]$$

$$\frac{-\text{Cov}(1+r_z, \Delta u'(c_2^C(r_z)))}{\mathbb{E}[1+r_z]} = \mathbb{E} [\Delta u'(c_2^C(r_z))] - \frac{\mathbb{E} [(1+r_z) \Delta u'(c_2^C(r_z))]}{\mathbb{E}[1+r_z]}.$$

Therefore

$$\Phi(\varphi) = \frac{\mathbb{E} [(1+r_z) \Delta u'(c_2^C(r_z))]}{\mathbb{E}[1+r_z]} - \mathbb{E} [\Delta u'(c_2^B(r_z))]$$

$$\Phi(\varphi) = \frac{\mathbb{E} [(1+r_z) (u'(c_2^{C1}(r_z)) - u'(c_2^{C0}(r_z)))]}{\mathbb{E}[1+r_z]} - \mathbb{E} [u'(c_2^{B1}(r_z)) - u'(c_2^{B0}(r_z))]$$

Now, taking the derivative

$$\Phi'(\varphi) = \frac{\mathbb{E} \left[(1+r_z) \left(u''(c_2^{C1}(r_z)) \frac{\partial c_2^{C1}}{\partial \varphi} - u''(c_2^{C0}(r_z)) \frac{\partial c_2^{C0}}{\partial \varphi} \right) \right]}{\mathbb{E}[1+r_z]} - \mathbb{E} \left[u''(c_2^{B1}(r_z)) \frac{\partial c_2^{B1}}{\partial \varphi} - u''(c_2^{B0}(r_z)) \frac{\partial c_2^{B0}}{\partial \varphi} \right]$$

Looking at the consumption derivatives, we have $\frac{\partial c_2^{B1}}{\partial \varphi} = \frac{\mathbb{E}[1+r_z]}{1-\nu} + (1+r_z) \frac{\partial s^{B1}}{\partial \varphi}$, $\frac{\partial c_2^{B0}}{\partial \varphi} = \frac{\mathbb{E}[1+r_z]}{1-\nu} + (1+r_z) \frac{\partial s^{B0}}{\partial \varphi}$, $\frac{\partial c_2^{C1}}{\partial \varphi} = (1+r_z) \frac{\partial s^{C1}}{\partial \varphi}$, and $\frac{\partial c_2^{C0}}{\partial \varphi} = (1+r_z) \frac{\partial s^{C0}}{\partial \varphi}$.

$$\begin{aligned} \Phi'(\varphi) &= \frac{\mathbb{E} \left[(1+r_z)^2 \left(u''(c_2^{C1}(r_z)) \frac{\partial s^{C1}}{\partial \varphi} - u''(c_2^{C0}(r_z)) \frac{\partial s^{C0}}{\partial \varphi} \right) \right]}{\mathbb{E}[1+r_z]} \\ &\quad - \mathbb{E} \left[u''(c_2^{B1}(r_z)) \left[\frac{\mathbb{E}[1+r_z]}{1-\nu} + (1+r_z) \frac{\partial s^{B1}}{\partial \varphi} \right] - u''(c_2^{B0}(r_z)) \left[\frac{\mathbb{E}[1+r_z]}{1-\nu} + (1+r_z) \frac{\partial s^{B0}}{\partial \varphi} \right] \right]. \end{aligned}$$

With CRRA utility $u''(c) = -\theta c^{-\theta-1}$ for $\theta > 1$. Additionally, from Proposition 1, $\frac{\partial s^{CR}}{\partial \varphi} >$

$\frac{\partial s^{BR}}{\partial \varphi}$. Since $u''(c) < 0$ and $\frac{\partial s}{\partial \varphi} > 0$, each bracketed term in $\Phi'(\varphi)$ is negative. Let A denote the DC term and D the DB term, so that $\Phi'(\varphi) = A - D$. Both A and D are negative, but the DB term is larger in absolute value because it contains the constant multiplier $\frac{\mathbb{E}[1+r_z]}{1-\nu}$ in addition to the savings-response component. Hence $D < A$ and $\Phi'(\varphi) = A - D > 0$. Therefore, $\Phi'(\varphi) > 0$ for all φ , so $\Phi(\varphi)$ is strictly increasing. Since $\Phi(\varphi)$ is strictly increasing in φ , φ_I^* is unique. $\square$

B Additional HRS Data Details

B.1 Sample Selection

The original HRS dataset includes 280,343 unique observations of 42,405 individuals. We exclude participants missing key demographic information (age, race, or region), those already retired at their first interview, and those lacking valid data on labor force status, retirement plans, retirement status, reported health status, union coverage, or health, long-term care, and life insurance coverage. Table 13 summarizes the sample size changes at each step.

Table 13: Sample Selection from HRS, 1992 to 2020

Sample Selection	# of Obs.	# of Ind.
Original panel	280343	42405
Drop participants without age info	280341	42404
Drop participants without race info	279908	42290
Drop participants without location info	279883	42281
Drop participants that already retired by their first interview	237015	33271
Drop obs without labor force participation info	236358	33230
Drop obs if pension information is missing	235468	33173
Drop obs without retirement info	234801	33098
Drop obs without reported health info	234799	33096
Drop obs without union coverage info	233095	33036
Drop obs without health/long-term care/life insurance info	219887	32791

B.2 Additional Sample Descriptions

Table 14 summarizes the demographic composition of participants in the sample using individual-level survey weights. The sample contains nearly equal proportions of male and female respondents, and the majority of participants are White or Caucasian. More than 78% of the sample has a high school degree or higher, while 26% have at least a bachelor's degree. Additionally, more than 92% of participants had been married at least once before exiting the study, highlighting the importance of accounting for family and household characteristics when analyzing how retirement plan choices affect savings and retirement decisions.

Table 14: Compositions of Participants in Sample

Characteristics	Proportion
Gender	
Male	49.00%
Female	51.00%
Race & Ethnicity	
White/Caucasian	79.12%
Black/African American	11.72%
Other	9.15%
Highest Degree	
No Degree	16.62%
GED	4.60%
HS	26.89%
HS/GED	18.92%
AA	6.79%
BA	16.59%
MA/MBA	7.99%
Law/MD/PhD	1.58%
Marital History	
Ever Married	92.51%
Never Married	7.49%

We compute the distribution of participants by the total number of observed retirement plan switches. The results indicate that the vast majority of participants (85.82 percent) never change their retirement plan type over the sample period. Approximately 8.21 percent switch once, 4.04 percent switch twice, and only about 2 percent experience three or more switches. These patterns suggest that retirement plan type is highly persistent, with most individuals maintaining the same plan type throughout their main job.

C Further Empirical Results

C.1 Robustness Check: Differences in Retirement Timing by Plan Types

This section presents additional robustness checks for the empirical relationship between retirement plan type and retirement timing. We first examine whether the estimated effect of pension plan type varies across generations by interacting plan type indicators with five-year birth cohort fixed effects. Model (4) in Table 15 reports the results from this specification.

When allowing the DC effect to vary across cohorts, the coefficient on DC reflects the effect

for the omitted cohort rather than the average effect across all cohorts. The relevant test is the joint significance of the plan-by-cohort interaction terms. The DC-by-cohort interactions are jointly insignificant ($\chi^2 = 4.51$, $p = 0.607$), and the Hybrid-by-cohort interactions are also jointly insignificant ($\chi^2 = 4.24$, $p = 0.515$). These results provide no evidence that the retirement-plan differential varies systematically across cohorts. Cohort-specific marginal effects for DC coverage are generally negative, consistent with the baseline finding that DC participants are less likely than DB participants to transition into retirement.

We next assess whether the estimates are driven by sectoral composition. Because public-sector jobs predominantly offer DB plans, retirement behavior in that sector may differ for reasons unrelated to plan design. To address this concern, we re-estimate the baseline model after excluding individuals employed in Public Administration. Model (5) in Table 15 reports the results for employment-to-retirement transitions among participants aged 60 to 69. The estimated plan effects remain similar in magnitude and statistical significance to those reported in Table 2, indicating that our main findings are not driven by the presence of government workers in the sample.

Because the hybrid-plan estimates are insignificant, we do not emphasize them in the baseline analysis and instead focus on the comparison between DB and DC plan enrollees. We estimate the differential in final retirement ages between workers whose primary retirement plan is a DB plan and those whose primary plan is a DC plan. Since retirement plan enrollment may be endogenous, we supplement the OLS estimates with an instrumental-variables specification that uses pre-retirement union coverage as an instrument for DC enrollment. The key concern is that workers may sort into jobs with different pension structures based on unobserved preferences for job stability, career attachment, or retirement timing, which would bias the OLS estimates. Columns (1) and (2) of Table 16 compare the OLS and IV estimates. The OLS estimate suggests that DC enrollment is associated with a 0.6-year increase in final retirement age relative to DB enrollment. In contrast, the IV estimate implies a substantially larger effect of approximately 5.2 years. This difference suggests that the OLS estimate may understate the effect of retirement plan type on retirement timing. One possible explanation is negative selection into DC plans: workers with weaker attachment to their employer, less stable career trajectories, or stronger preferences for earlier retirement may be more likely to hold DC plans, thereby attenuating the OLS estimate. Column (3) excludes public-sector workers, and the estimated DC-DB retirement-age differential remains very similar, suggesting that the IV result is not driven by public-sector employment.

Union coverage strongly predicts retirement plan type in the first stage. The coefficient on pre-retirement union status is -0.25 and it is statistically significant at 1% level ($t = -7.07$), indicating that unionized workers are substantially less likely to be enrolled in DC plans. The excluded-instrument F-statistic is 49.97, well above conventional weak-instrument thresholds (Stock-Yogo 10% critical value = 16.38). This indicates that the instrument is highly relevant and that weak identification is unlikely to be a concern.

Because the pooled IV estimate is based on the full 1992-2020 sample, it may mask changes in the DB-DC retirement-age differential over time. We therefore examine whether the estimated

gap differs between the earlier and later parts of the sample. Table 17 reports IV-based predicted retirement ages by plan type for two broad periods. Because the final-retirement-age sample is relatively small, we examine changes in the DB-DC retirement-age differential using two broad periods, 1992-2008 and 2010-2020. DC participants have higher predicted retirement ages than DB participants in both periods. The estimated DC-DB differential rises from 4.545 years in 1992-2008 to 6.071 years in 2010-2020. However, the increase of 1.526 years is not statistically significant ($\chi^2_1 = 0.33$, $p = 0.565$). Therefore, we find no evidence that the DB-DC retirement-age gap changed significantly over time.

We exclude workers without a retirement plan from the baseline DB-DC comparison because the main empirical analysis focuses on differences between traditional DB and DC coverage. However, for completeness, and to assess whether the DB-DC pattern is sensitive to the treatment of other plan categories, we also estimate employment-to-retirement transition probabilities using a broader four-category retirement-plan classification: no plan, DB, DC, and hybrid.

Table 18 reports the results under alternative fixed-effect specifications. Across all specifications, DC coverage is associated with a significantly lower probability of transitioning from employment to retirement relative to DB coverage. The estimated DC coefficients range from -0.424 to -0.439 and remain statistically significant at the 1% level, indicating that the baseline DB-DC pattern is robust to including workers without retirement plans. Workers without a retirement plan are also significantly less likely to transition into retirement than DB participants, with estimated coefficients around -0.50 . This pattern is consistent with the interpretation that workers lacking retirement benefits may delay retirement because they have fewer retirement resources. In contrast, the coefficients on hybrid plans are small and statistically insignificant, suggesting no clear difference in retirement transitions between hybrid-plan and DB-plan participants in this sample. Overall, the four-plan specification confirms the main empirical finding: relative to DB participants, DC participants are less likely to retire from employment during ages 60–69. The inclusion of workers without retirement plans does not alter this qualitative conclusion.

Table 19 shows the predicted retirement ages using a broader four-category retirement-plan classification. The first specification includes four plan categories: no plan, DB, DC, and hybrid.³³ In this specification, workers without an employer-sponsored retirement plan have the highest predicted retirement age, 69.910, followed by hybrid-plan participants, DC participants, and DB participants. The high predicted retirement age among workers without a plan is consistent with the idea that limited employer-sponsored retirement resources may

³³Since this specification includes indicators for more than two retirement plan categories, we rely on a broader set of instruments. In particular, we use both union status and pre-retirement job type (industry, occupation) as instruments for retirement plan enrollment. For occupation and industry to serve as valid instruments, the identifying assumption requires that they influence retirement age only through their effects on retirement plan type and not through any independent channel. This assumption is plausible for two reasons. First, the regression includes controls for factors that could mediate the relationship between occupation or industry and retirement incentives, including self-reported health status, which captures variation in physical capacity and health-related motives for early or delayed retirement. Second, by conditioning on union status and retirement plan type, the specification absorbs the institutional and financial incentives that differ across jobs and sectors.

require workers to remain employed longer to finance retirement. However, this estimate should be interpreted cautiously because workers without retirement plans may differ from pension-covered workers in ways not fully captured by the controls.

Including the no-plan group reduces the estimated DB-DC differential. In the four-category specification, the predicted retirement age is 61.201 for DB participants and 63.263 for DC participants, implying a DC-DB gap of about 2.1 years. When the sample is restricted to DB, DC, and hybrid plans, the corresponding predicted ages are 61.393 for DB participants and 66.367 for DC participants, implying a larger DC-DB gap of about 5.0 years. However, the qualitative pattern remains the same: DC participants have higher predicted retirement ages than DB participants. The predicted retirement age for hybrid-plan participants is high in both specifications, but the estimate is imprecise, reflecting the small number of hybrid-plan observations in the HRS. For this reason, we do not emphasize the hybrid-plan estimates in the baseline analysis.

Table 15: *Employment-to-Retirement Transitions Among Ages 60-69 (Robustness Check)*

	(4) Plan $\times$ Cohort		(5) Exclude Gov Workers	
	Estimates	S.E.	Estimates	S.E.
Retirement Plans				
DC	0.479	1.190	-0.437***	0.065
Hybrid	-0.241	0.826	-0.065	0.141
Retirement Plans $\times$ Cohort Bin				
DC $\times$ 1925-1929	-0.393	1.249		
DC $\times$ 1930-1934	-0.882	1.198		
DC $\times$ 1935-1939	-0.893	1.195		
DC $\times$ 1940-1944	-1.028	1.196		
DC $\times$ 1945-1949	-0.756	1.202		
DC $\times$ 1950-1954	-1.007	1.214		
Hybrid $\times$ 1925-1929	-0.339	1.411		
Hybrid $\times$ 1930-1934	0.452	0.871		
Hybrid $\times$ 1935-1939	-0.031	0.855		
Hybrid $\times$ 1940-1944	0.043	0.866		
Hybrid $\times$ 1945-1949	0.772	0.933		
Demographic Controls				
Age	3.434***	0.456	3.332***	0.471
Age squared	-0.027***	0.004	-0.026***	0.004
Female	0.174*	0.072	0.202**	0.076
Never married	-1.283	1.071	-1.275	1.072
Race: Black/African American	-0.216*	0.101	-0.214*	0.107
Race: Other	-0.324*	0.156	-0.383*	0.164
Income Controls				
Hourly wage	0.004*	0.002	0.005*	0.002
Hourly wage squared	-0.000	0.000	-0.000	0.000
Income from retirement plans	0.000*	0.000	0.000**	0.000
Income from retirement plans squared	-0.000	0.000	-0.000	0.000
Income from social security	0.000	0.000	0.000	0.000
Social security income squared	-0.000	0.000	-0.000	0.000
Household wealth (incl. house)	-0.000	0.000	-0.000	0.000
Health Controls				
Self-reported health: very good	0.180*	0.088	0.184*	0.092
Self-reported health: good	0.395***	0.091	0.392***	0.094
Self-reported health: fair	0.527***	0.117	0.534***	0.121
Self-reported health: poor	0.542*	0.230	0.453	0.245
Spouse health: very good	0.050	0.091	0.049	0.094
Spouse health: good	-0.050	0.093	-0.054	0.097
Spouse health: fair	-0.171	0.111	-0.164	0.115
Spouse health: poor	0.132	0.146	0.156	0.151
Insurance Coverage Controls				
Health insurance coverage	0.177	0.253	0.189	0.256
Long-term care insurance coverage	0.140	0.091	0.140	0.095
Life insurance coverage	-0.038	0.092	0.003	0.095
Constant	-113.730***	14.399	-109.447***	14.842
Observations	7,693		7,163	

Notes: This table reports logit estimates of employment-to-retirement transitions among individuals aged 60–69. Model (4) includes interactions between retirement plan type and birth cohort. Model (5) excludes individuals employed in Public Administration. All specifications are estimated using HRS individual-level weights and include controls for education, occupation, industry, region, birth cohort, and year fixed effects. Robust standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Estimated DC–DB Retirement Age Differential

	(1) OLS	(2) IV	(3) IV (Exclude Gov Workers)
DC	0.612* (0.373)	5.239*** (1.395)	5.116*** (1.463)
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,035	1,035	933
First-stage F	–	49.97	45.87

Notes: The table reports the estimated retirement age differential between DC and DB plans (DB is the reference category) from Equation (1). Columns report OLS, IV, and IV excluding public-sector workers. Robust standard errors are reported in parentheses. Statistical significance is indicated by asterisks: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Predicted Retirement Age by Retirement Plan Type and Period

	1992–2008	2010–2020
<i>Panel A. Predicted retirement age</i>		
DB	60.587*** (0.465)	62.759*** (1.250)
DC	65.281*** (1.157)	69.044*** (1.212)
<i>Panel B. DC-DB retirement-age differential</i>		
DC-DB	4.545***	6.071***
Difference across periods		1.526
χ^2_1		0.33
p -value		0.565

Notes: Panel A reports IV-based predicted final retirement ages by retirement plan type and period. Robust standard errors are reported in parentheses, and 95% confidence intervals are reported in brackets. Panel B reports the estimated DC-DB retirement-age differential within each period and tests whether the differential differs between the two periods. The difference across periods is defined as the 2010-2020 DC-DB differential minus the 1992-2008 DC-DB differential. Statistical significance is indicated by asterisks: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 18: Employment to Retirement Transitions Among Ages 60-69

	(1) Year FE		(2) Cohort FE		(3) Year + Cohort FE	
	Estimates	S.E.	Estimates	S.E.	Estimates	S.E.
Retirement Plans						
No Plan	-0.498***	0.060	-0.496***	0.060	-0.493***	0.060
DC	-0.435***	0.060	-0.424***	0.060	-0.439***	0.061
Hybrid	-0.054	0.133	-0.088	0.133	-0.074	0.133
Demographic Controls						
Age	1.973***	0.295	1.954***	0.297	1.949***	0.299
Age squared	-0.015***	0.002	-0.015***	0.002	-0.015***	0.002
Female	0.104	0.056	0.088	0.056	0.095	0.056
Never married	-0.323	0.462	-0.397	0.465	-0.401	0.463
Race: Black/African American	-0.238**	0.075	-0.259***	0.075	-0.262***	0.075
Race: Other	-0.306**	0.119	-0.296*	0.119	-0.304*	0.120
Income Controls						
Hourly wage	-0.001	0.001	-0.001	0.001	-0.001	0.001
Hourly wage squared	0.000	0.000	0.000	0.000	0.000	0.000
Income from retirement plans	0.000	0.000	0.000	0.000	0.000	0.000
Income from retirement plans squared	0.000	0.000	0.000	0.000	0.000	0.000
Income from social security	0.000	0.000	0.000	0.000	0.000	0.000
Social security income squared	-0.000	0.000	-0.000	0.000	-0.000	0.000
Household wealth (incl. house)	-0.000	0.000	-0.000	0.000	-0.000	0.000
Health Controls						
Self-reported health: very good	0.189**	0.069	0.191**	0.069	0.187**	0.069
Self-reported health: good	0.349***	0.071	0.349***	0.071	0.353***	0.071
Self-reported health: fair	0.489***	0.087	0.497***	0.087	0.490***	0.087
Self-reported health: poor	0.481**	0.168	0.505**	0.168	0.478**	0.168
Spouse health: very good	0.137	0.071	0.138	0.071	0.131	0.071
Spouse health: good	-0.020	0.073	-0.020	0.073	-0.025	0.073
Spouse health: fair	0.016	0.084	0.015	0.084	0.007	0.084
Spouse health: poor	0.198	0.106	0.204	0.106	0.196	0.107
Insurance Coverage Controls						
Health insurance coverage	0.010	0.110	0.011	0.110	0.013	0.111
Long-term care insurance coverage	0.102	0.072	0.092	0.073	0.103	0.073
Life insurance coverage	-0.025	0.060	-0.023	0.060	-0.022	0.061
Constant	-66.450***	9.359	-64.907***	9.470	-64.759***	9.536
Observations	13,163		13,232		13,153	

Notes: The table shows the results of a logistic regression using HRS sample. Estimations are adjusted by the HRS personal-level weights. The regression also controls the participant's educational degree, occupation, industry, as well as region and year fixed effects. The reference category is married White men enrolled in a DB retirement plan in their current main job, with excellent self-reported health, no health insurance, no long-term care insurance, and no life insurance. Their spouses also report excellent health. Robust standard errors are presented. Stars denote statistical significance * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 19: Predicted Retirement Age by Retirement Plan Type

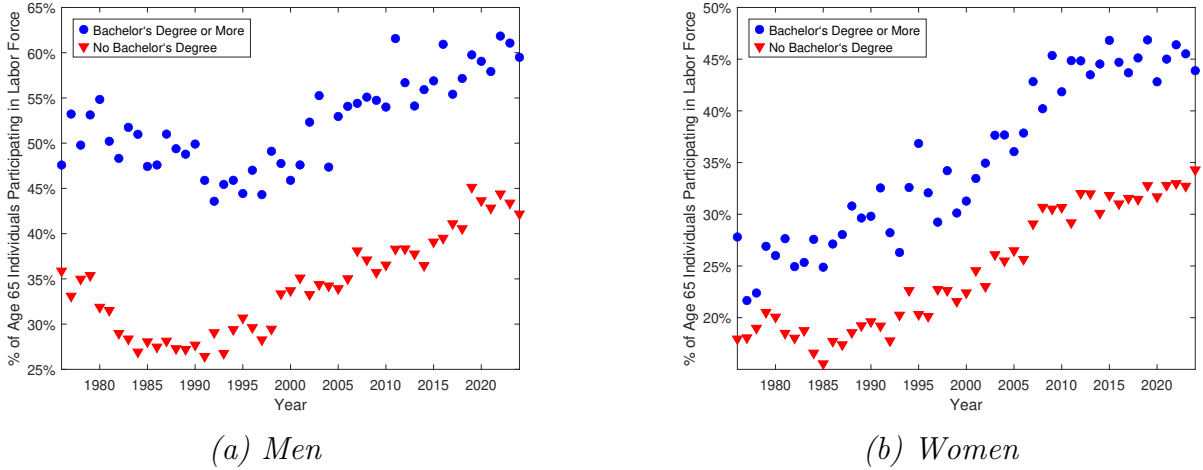
	Four Plan Categories	Three Plan Categories
No Plan	69.910*** (0.915)	—
DB	61.201*** (0.500)	61.393*** (0.406)
DC	63.263*** (1.368)	66.367*** (0.670)
Hybrid	70.503*** (6.750)	69.030*** (4.789)

Notes: The table reports IV-based predicted final retirement ages by retirement plan type. The four-category specification includes individuals with no employer-sponsored retirement plan, DB plans, DC plans, and hybrid plans. The three-category specification restricts the sample to individuals with DB, DC, or hybrid plans. Robust standard errors are reported in parentheses. Statistical significance is indicated by asterisks: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Decline in Retirement at Age 65

Figure 2 in the main text shows that the participation rate among age 65 individuals has been rising since the 1990s in the United States. Could some of this increase be due to differences in educational attainment during that time-frame, with more educated individuals tending to participate in the labor force later into the life-cycle? The following figure shows that the trend displayed in Figure 2 is robust when separately looking at men and women with different levels of educational attainment.

Figure 9: Labor Force Participation at Age 65 by Education and Gender



C.3 Empirical Evidence Related to Changes in Right-to-Work (RTW) Laws

This section applies a difference-in-differences (DID) approach to estimate the impact of the passage of the RTW laws, which is associated with a cut in union coverage (Fortin, Lemieux and Lloyd, 2022), on the coverage of DB plans. As mentioned in Section 3, this exercise provides additional evidence that union status is a strong instrument for the retirement plan type when estimating the Equation (1). As illustrated in Table 20, RTW laws were adopted by U.S. states in a staggered fashion over several decades. Recent econometric literature highlights that traditional Two-Way Fixed Effects (TWFE) estimates can be biased in the presence of staggered treatment timing and heterogeneous treatment effects (Callaway and Sant'Anna (2021)). To address the concerns, this exercise applies staggered DID by Callaway and Sant'Anna (2021).

$$Y_{i,s,t} = \beta RTW_{s,t} + \mathbf{X}_{i,s,t} + \delta_t + \alpha_s + \epsilon_{i,s,t} \quad (23)$$

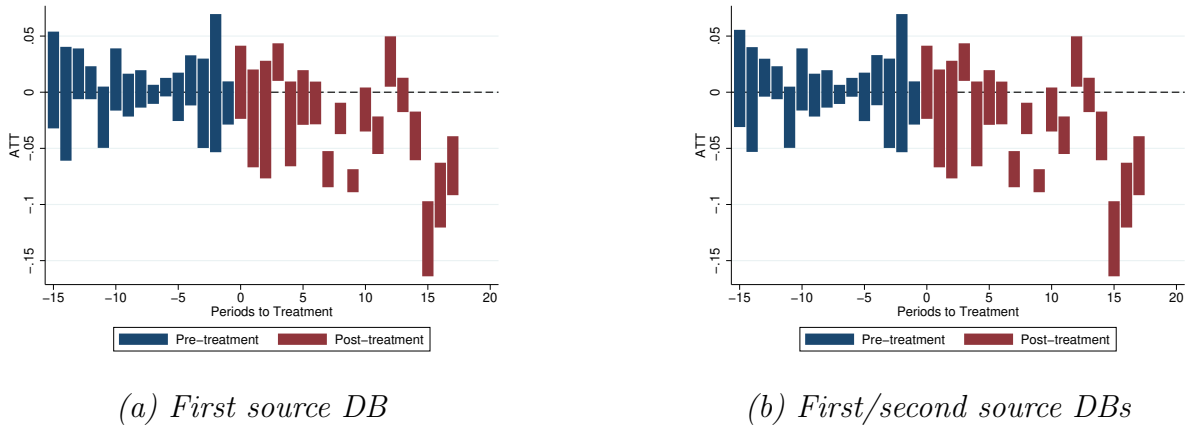
The empirical specification is shown in the Equation (23). $Y_{i,s,t}$ is a binary variable for whether an individual's retirement income source is DB plan. The treatment variable $RTW_{s,t}$ is defined by the year of RTW adoption. $\mathbf{X}_{i,s,t}$ includes person-level controls for age, gender, and race. Following standard practice, standard errors are clustered at the state level to account for within-state serial correlation.

Table 20: RTW Laws Adoption Timetable in the U.S.

State	Year	State	Year
Alabama	1953	Nevada	1952
Arizona	1947	North Carolina	1947
Arkansas	1944	North Dakota	1948
Florida	1944	Oklahoma	2001
Georgia	1947	South Carolina	1954
Idaho	1985	South Dakota	1947
Indiana	2012	Tennessee	1947
Iowa	1947	Texas	1947
Kansas	1958	Utah	1955
Kentucky	2017	Virginia	1947
Louisiana	1976	Wisconsin	2015
Michigan	2012	West Virginia	2016
Mississippi	1954	Wyoming	1963
Nebraska	1946		

Data source: National Right to Work Committee website and state government websites.

Figure 10: Dynamic Treatment Effects Plot



Notes: Data source is CPS Annual Social Economic Supplement (CPS-ASEC). Standard errors are clustered at the state level. Estimations are adjusted by the ASEC weights. The sample age is from 18 to 75. The sample period is from 1977 to 2024. (a) shows the results when the first source of retirement income is from DB plan. While (b) shows the results when either the first or the second source is from DB plan. DB plan includes Company or Union pension, Federal Government retirement Pension, US Military retirement pension, State or local Gov't retirement pension, or US Railroad retirement pension.

Figure 10 presents the dynamic event study plot. Visual inspection reveals that the pre-treatment effect is flat and around zeros. Following the adoption of RTW laws, it takes about 15 years before the effects on DB plan inclusion get pronounced. The delayed impact arises because the shift in retirement plan types is less likely to affect current employees than it is to affect new hires.

The aggregated average treatment effects are reported in Table 21. The average pre-treatment effects are close to zero (0.0008 for first source DB; 0.0009 for first or second source DB) and statistically insignificant, further supporting the validity of parallel trend assumption. Conversely, the post-treatment averages indicate that RTW laws adoptions

significantly reduce the likelihood of DB plan coverage by 3.23 percentage points.

Table 21: Aggregated Treatment Effects of RTW Laws on DB plan inclusion

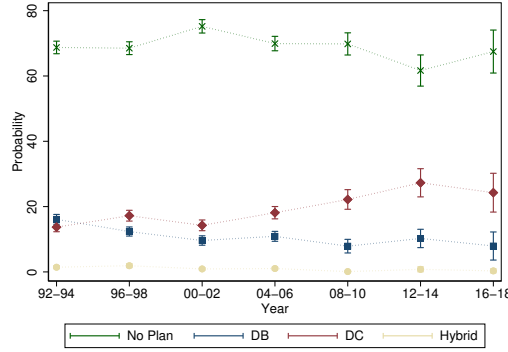
	(1)	(2)
	First source DB	First/second source DB
Pre-Treatment Average	0.0008 (0.0016)	0.0009 (0.0016)
Post-Treatment Average	-0.0323*** (0.0062)	-0.0323*** (0.0062)
Observations	136,392	136,392

*Notes: Data source is CPS-ASEC. Standard errors are clustered at the state level. Stars denote statistical significance * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Estimations are adjusted by the ASEC weights. The sample age is from 18 to 75. The sample period is from 1977 to 2024. (1) shows the results when the first source of retirement income is from DB plan. While (2) shows the results when either the first or the second source is from DB plan. DB plan includes Company or Union pension, Federal Government retirement Pension, US Military retirement pension, State or local Gov't retirement pension, or US Railroad retirement pension.*

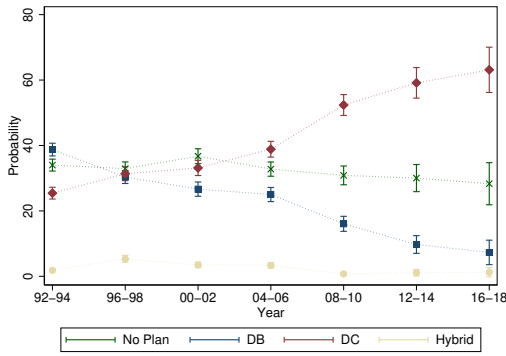
C.4 Plan Enrollment Probabilities by Years and Earning Percentiles

To examine how the likelihood of enrolling in different types of retirement plans varies across the earnings distribution and over time, we estimate a multinomial logit model of retirement plan enrollment. We begin by grouping survey waves into seven two-year periods from 1992 to 2018 and constructing earnings quantiles separately within each year to ensure that the quantile assignments are not influenced by changes in the aggregate earnings distribution over time. Participants are classified into three earnings quantiles within each survey year based on their real labor income. We then estimate a multinomial logit specification in which the dependent variable is the individual's primary retirement plan type (no plan, defined benefit, defined contribution, or hybrid). The model includes age, earnings quantile, a full set of interactions between earnings quantile and year group, the real hourly wage, and demographic characteristics such as gender, race, education, marital status, and region. We also control for occupation, industry, and year-group fixed effects, and all estimations apply the personal-level sampling weights from the HRS. Figure 11 plots the predicted probabilities of enrolling in each of the four retirement plan types over time for individuals in the three earnings quantiles.

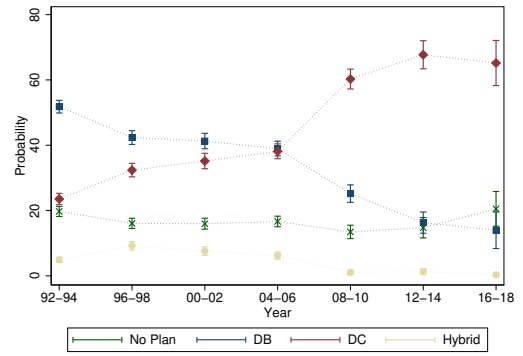
Figure 11: Changes in Retirement Plan Enrollment Probability: HRS Data



(a) 25th Percentile Earnings



(b) 50th Percentile Earnings



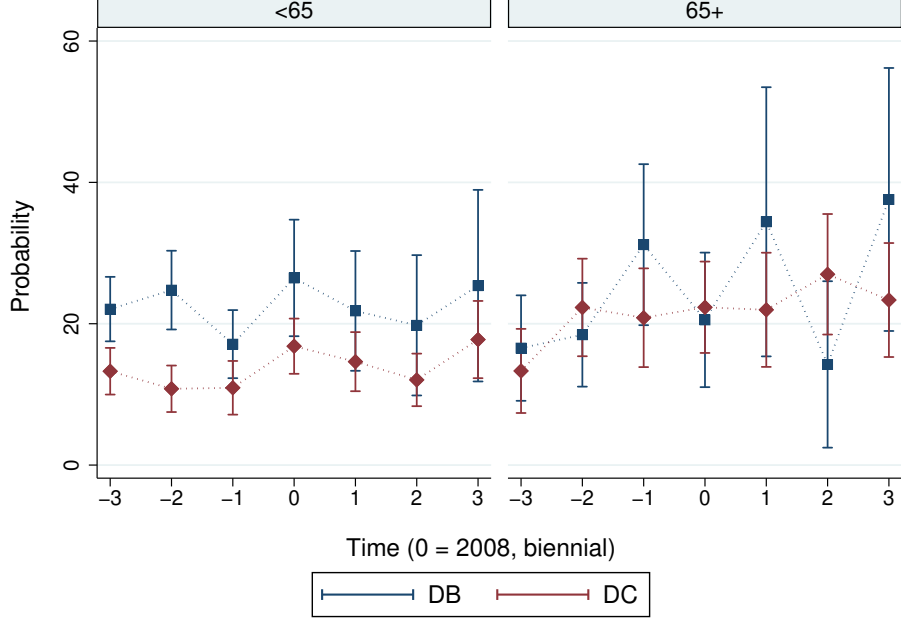
(c) 75th Percentile Earnings

C.5 Effect of Risky Returns on Retirement Decisions by Retirement Plans

A related question is whether individuals enrolled in DC plans respond differently to fluctuations in the market rate of return than those enrolled in DB plans. To investigate this, we focus on years in which investment returns were very low or negative, such as 2008, and examine whether the probability of transitioning from employment to retirement changes differentially for DC participants relative to DB participants. Consistent with the earlier analysis, we restrict the sample to individuals between ages 50 and 70. We estimate a logistic regression in which the dependent variable is an indicator for transitioning from employment to retirement between survey waves. The key explanatory variables are a set of event-time indicators around the 2008 crisis year, retirement plan type dummies (DB and DC), and age-group indicators (younger than 65 and 65 or older). We include the full set of interactions among these variables, which allows the effect of the crisis year to vary jointly by plan type and age group. The regression additionally controls for participants' earnings, household's wealth, self-reported health, gender, race, educational attainment, marital status, occupation, industry, and region, and applies personal-level sampling weights. Standard errors are clustered at the individual level to account for

repeated observations within persons.

Figure 12: Predicted Employment to Retirement Probability Around 2008 Recession by Plans



Using the estimated coefficients from this model, we compute the predicted probability of a transition from employment to retirement by retirement plan type and event time, separately for each age group. Figure 12 shows no statistically significant differences in the response of employment-to-retirement transitions across plan types for all ages.

D Model Extensions

This appendix considers two extensions of the quantitative model in the main text, each relaxing one baseline assumption at a time. The first allows retired agents to reenter the labor force. The second allows Social Security benefits to depend on retirement age. These exercises assess whether the central qualitative result of the model is sensitive to alternative assumptions governing retirement behavior. In both extensions, a shift from DB toward DC plans continues to reduce retirement at older ages and lower welfare, although the quantitative magnitudes and the timing of asset accumulation differ from those in the baseline model.

We do not repeat the full decomposition of the DB to DC transition presented in Table 11 for these extensions. The purpose of these exercises is to assess whether the central qualitative effects of the transition are sensitive to alternative retirement environments, rather than to reestimate the quantitative contribution of each mechanism. Allowing

labor force reentry or making Social Security benefits depend on retirement age changes the retirement incentives available to households, but neither extension changes the underlying differences in return and longevity risk across DB and DC plans. We therefore conduct the mechanism decomposition using the main quantitative model, which provides a more transparent environment for isolating the contributions of plan design, return risk, and longevity risk.

D.1 Labor Force Reentry

In the main text model, agents cannot reenter the labor force once they choose to retire. However, in reality, people can return to work after retirement. The model extension in this section allows for labor force reentry while keeping other assumptions the same as the quantitative model in the main text.

D.1.1 Value Functions

In this setup, retired agents can choose to stay retired ($R = 1$) or reenter the labor market ($R = 0$). Once reentering, the households face the same job finding rate as the unemployed. Meanwhile, those rejoining the labor force are subject to a one-time entry cost, c_e . The updated value function for the retired is changed as follows.

$$\begin{aligned} V_a^R(y, b, \gamma_B) = & \max_{s, R \in \{0,1\}} \{u(c, \ell_R) - (1 - R)c_e + \nu_a \beta \mathbb{E}[\zeta(b'_R(s), y, \gamma_B)] + \\ & R(1 - \nu_a) \beta \mathbb{E}[V_{a+1}^R(y, b'_R(s), \gamma_B)] + \\ & (1 - R)(1 - \nu_a) \beta (1 - \eta_{a+1}) \mathbb{E}[V_{a+1}^U(y, b'_R(s), \gamma_B)] + \\ & (1 - R)(1 - \nu_a) \beta \eta_{a+1} \sum_{i_p \in P} p_i(i_p | y) \mathbb{E}[V_{a+1}^E(y, b'_R(s), i_p, \gamma_B)]\} \end{aligned}$$

D.1.2 Calibration

The jointly calibrated parameters shown in Table 5 are recalibrated with the new assumption of reentry. In addition, the newly added one-time entry cost c_e is also jointly calibrated. This parameter is calibrated to match the percent of retired people who reentered the labor force between ages 50 and 60 in 1994-1999³⁴, estimated from the CPS data.

D.1.3 Key Result Replications

As in the baseline model, this model extension can also match the percent of retired agents by age reasonably well (see Figure 13). Different from Figure 7, as the retired can

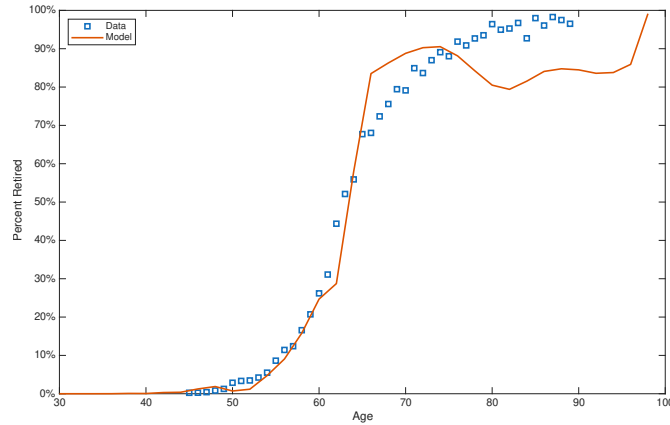
³⁴Prior to 1994, there was no information available regarding whether the individuals were not in the labor force due to retirement or not.

Table 22: Jointly Calibrated Parameter Values

Parameter	Estimate	Targeted Moment	Data	Model
σ_ϵ	0.0239	Quarterly returns coefficient of variation	2.66	2.70
ℓ_R	0.8750	Percent retired if age 65+	84.53%	85.47%
ζ_c	4.6700	% Change median net worth 55–64 to 75+	-23.64%	-22.73%
θ	1.3808	Age 55–64 median net worth to income ratio	4.81	4.41
χ	0.0297	Percent retired: age 70–74 to 65–69	1.17	1.07
c_e	2.1594	Percent of reentry: age 50–60	7.22%	6.76%

reenter the labor force, a noticeable share of the retired go back to work after age 70. In addition, the reentry assumption generates earlier retirement among individuals in their early 50s as observed in the data.

Figure 13: Percent Retired by Age (Untargeted)



Note: The data displayed in this figure is from the CPS in 1994, as we calibrate the model to the early 1990s. Prior to 1994, there was no information available regarding whether individuals were not in the labor force due to retirement or due to another reason.

Table 23 reports the estimated effects of the counterfactual movement from DB to DC plans. Consistent with the results in the main text, when labor-market reentry is allowed, the percentage of agents retired at ages ranging from 65 to 70+ remains lower in the economy with higher DC plan enrollment. Moreover, the percentage of retired at age 70+ decreases further with the labor-market reentry extension, which is partially because of the observed reentry after age 70 in Figure 13. So, with the transition, households are more likely to rejoin the labor force after initial retirement and increase lifetime labor participation. This further shows that households respond to longevity and return risk with greater reliance on DC plans by working longer, even with the choice of labor force reentry.

Next, the change in average assets at age 55 is slightly lower following the shift to DC plans with this model extension. This is partly because a sizable fraction of people (about 10% before the plan shift) retire in their early 50s and begin to draw down retirement savings earlier. As more people are enrolled in a DC plan, the consumption of retirement

Table 23: Estimated Effects of Retirement Plan Transition

	All Plan Enrollment Changes 1992–2019	Only transition out of DB
% Retired at Age 65–69	-0.4pp	-0.5pp
% Retired at Age 65	-4.0pp	-4.2pp
% Retired at Age 70+	-6.8pp	-6.5pp
Average Age 55 Assets	-1.6%	-2.0%
Average Age 80 Assets	+28.1%	+29.4%
Average Age 95 Assets	+33.0%	+33.0%
Entrant Consumption Equivalent Variation	-1.8%	-1.9%

savings accelerates. However, since the retired individuals can return to the labor force, more agents choose to go back to work to accumulate more assets for retirement. As a result, the average assets at ages 80 and 95 increase substantially in this higher-DC-enrollment economy, which indicates the insurance response of accumulating more assets with greater reliance on DC plans. In addition, consistent with Table 9, the estimated CEV also decreases.

D.2 Age-Dependent Social Security Income

In the baseline quantitative model, retired agents receive a uniform Social Security benefit after age 65. This section extends the model by introducing a retirement-age-dependent benefit schedule and demonstrates that the main results remain robust to this alternative specification.

In this economy, retired agents start to receive Social Security benefits from age 62. The amount of the benefit depends on their retirement age. Agents can receive full retirement benefits if they retire at the full retirement age (FRA), which is 65³⁵. For early retirement, the benefit payment is reduced. For delayed retirement, agents receive additional credit for each extra month they work, up to age 69.

D.2.1 Value Functions

The value function for the retired now depends on both the current age, a , and the retirement age, a_r .

$$\begin{aligned}
 V_a^R(y, b, \gamma_B, a_r) = & \max_s \{u(c, l_R) + \nu_a \beta \mathbb{E}[\zeta(b'_R(s), y, \gamma_B)] \\
 & + (1 - \nu_a) \beta \mathbb{E}[V_{a+1}^R(y, b'_R(s), \gamma_B, a_r)]\}
 \end{aligned} \tag{24}$$

The Social Security income, $\kappa_a(a_r)$, also depends on the two states. For instance, if a

³⁵In 1983, Congress changed the full retirement age from 65 to 67. For people born after 1960, the FRA is 67. For people born between 1938 and 1959, the full retirement age increased in increments. For people born before 1938, the FRA is 65.

worker retires before age 62, the worker will not receive Social Security benefits until the age of 62. If a worker retires early, the Social Security payment is subject to an early-retirement penalty. The detailed Social Security benefit schedule is specified below:

$$\kappa_a(a_r) = \begin{cases} 0, & a < 62 \\ \kappa_a(1 - \Delta(a_r = 62)), & a \geq 62, a_r < 62 \\ \kappa_a(1 - \Delta(a_r)), & a \geq 62, 62 \leq a_r < 65 \\ \kappa_a(1 + \Delta(a_r)), & a \geq 62, 65 \leq a_r < 70 \\ \kappa_a(1 + \Delta(a_r = 69)), & a \geq 62, a_r \geq 70 \end{cases}$$

D.2.2 Calibration

The Social Security benefit schedule is calculated according to the Social Security Administration’s website³⁶. The annual average incremental income is shown in Figure 14. In addition, the jointly calibrated parameters are recalibrated in this model extension as presented in Table 24.

Table 24: Jointly Calibrated Parameter Values

Parameter	Estimate	Targeted Moment	Data	Model
σ_ϵ	0.0239	Quarterly returns coefficient of variation	2.66	2.70
ℓ_R	1.0469	Percent retired if age 65+	84.53%	95.57%
ζ_c	4.2481	% Change median net worth 55–64 to 75+	-23.64%	-23.19%
θ	0.9387	Age 55–64 median net worth to income ratio	4.81	4.62
χ	0.0378	Percent retired: age 70–74 to 65–69	1.17	1.20

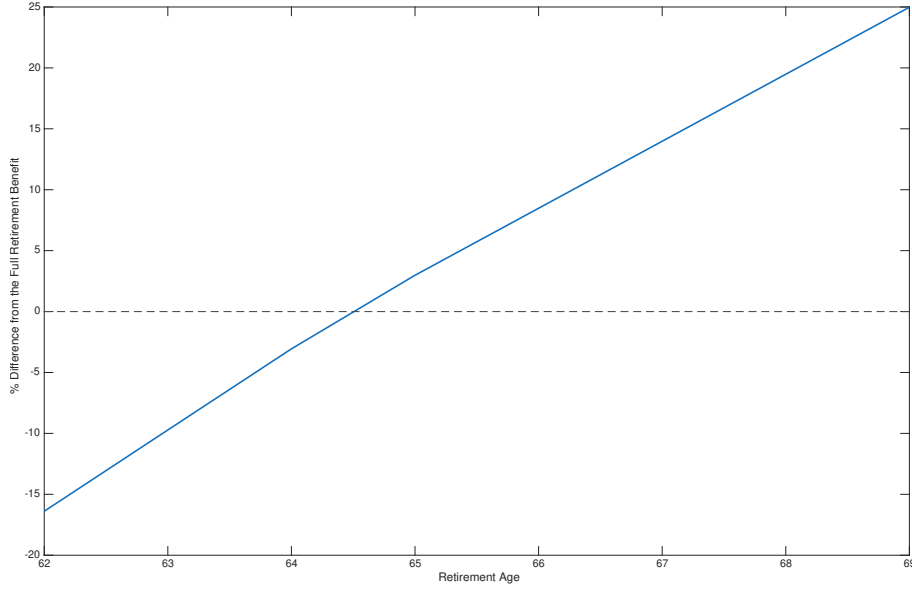
D.2.3 Key Result Replications

The untargeted percentage of retirees by age in the model still follows a pattern similar to that in the data (shown in Figure 15). Compared with the baseline model in the main text, a larger share of agents who retire early, particularly in their early 50s. This is due to the change in the Social Security payment schedule. Although agents who retire earlier face a penalty relative to the full retirement benefit, they also begin receiving Social Security income three years earlier than in the baseline model. In addition, the leisure from retirement, ℓ_R , is also higher in this model extension than that in the baseline model. Earlier retirement reflects both earlier access to Social Security benefits and the higher recalibrated utility value of retirement leisure.

Table 25 reports the counterfactual analysis results with the shift from DB to DC plans. Consistent with the findings in the main text, this shift decreases the share of agents who retire at ages 65 to 69. The share of agents who retire at age 70 and older

³⁶https://www.ssa.gov/OACT/quickcalc/early_late.html.

Figure 14: Social Security Income Schedule



Note: Annual average changes.

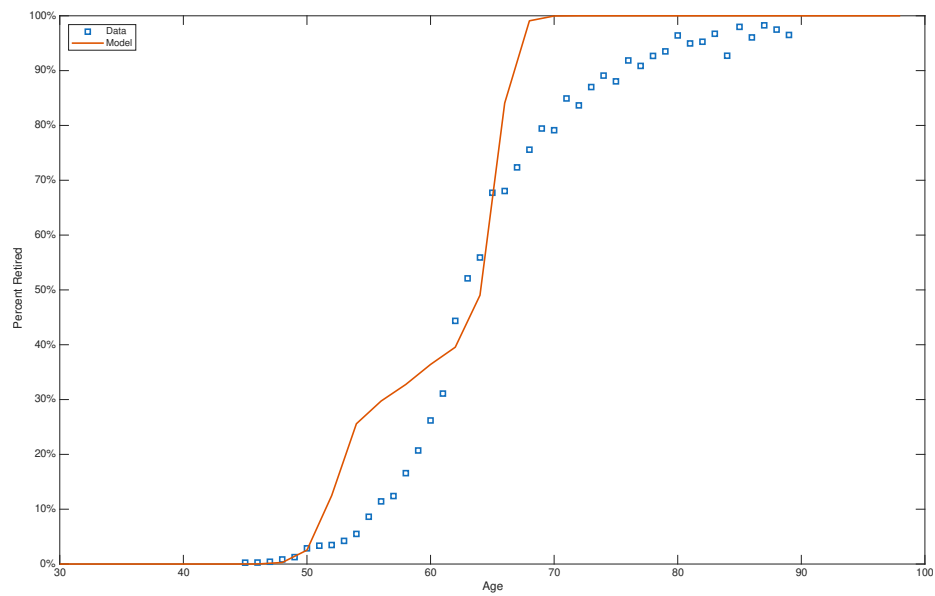
Table 25: Estimated Effects of Retirement Plan Transition

	All Plan Enrollment Changes 1992–2019	Only transition out of DB
% Retired at Age 65–69	-11.6pp	-11.5pp
% Retired at Age 65	-34.5pp	-34.5pp
% Retired at Age 70+	+0.0pp	+0.0pp
Average Age 55 Assets	-25.5%	-26.4%
Average Age 80 Assets	+15.3%	+15.1%
Average Age 95 Assets	+0.5%	+0.5%
Entrant Consumption Equivalent Variation	-2.2%	-2.3%

stays the same after the counterfactual, likely because nearly all agents in both economies have retired before age 70 with this model extension.

The average assets decrease at age 55 might result from retirement in the early 50s under this model extension. With the transition to an economy with more DC plans, this may reflect more people starting to draw savings after retirement in their early 50s. Average assets at age 80 remain higher after the DC transition, although the increase magnitude is smaller than that in the main text model. This might be because agents who delay retirement beyond age 65 receive additional Social Security payments, which partially substitute the private savings they need to accumulate for retirement. This indicates that people still insure themselves against longevity and return risk with more savings, with additional Social Security benefits for delayed retirement. Average assets at age 95 are essentially unchanged following the transition, with a slight increase of 0.5%. Regarding the estimated CEV, the last row of the table shows that welfare declines further after the DB-to-DC plan transition with the extended Social Security benefit schedule.

Figure 15: Percent Retired by Age (Untargeted)



Note: The data displayed in this figure is from the CPS in 1994, as we calibrate the model to the early 1990s. Prior to 1994, there was no information available regarding whether individuals were not in the labor force due to retirement or due to another reason.